

The Eyes of Texas are Upon OB/GYNs: Physician Migration and Crowdsourced Enforcement of Abortion Policy *

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Abstract

This paper studies the labor market effects of private liability enforcement using Texas Senate Bill 8 (SB-8, 2021) as a natural experiment. A private enforcement mechanism in this context allows any citizen to sue anyone who “aids or abets” a prohibited abortion for minimum damages of \$10,000. This “bounty law” lowers the evidentiary burden and expands the pool of potentially liable individuals, which raises expected operating costs and could incentivize out-migration for a broad set of physicians compared to traditional state enforcement. We exploit variation across physician specialties, states, and time in a synthetic difference-in-differences (SDID) design, comparing reproductive health physicians to all other physicians in Texas against similarly regulated states before and after SB-8’s signing. Using a Medicare administrative provider panel of 1,639,519 physicians (2015–2022) and Wagescape job posting data (2016–2022), we find that reproductive health physicians did not leave Texas at differential rates following SB-8: our SDID estimate allows us to rule out an effect larger than 989 physicians moving, a 4.1% change relative to the pre-period baseline, ruling out effects as large as those found in most of the contemporary literature on physician migration responses to state-enforced abortion restrictions. The null result is consistent with the high fixed cost of interstate relocation, estimated at \$150,000 to \$250,000. Neither posted salaries nor job posting volume for reproductive health providers changed differentially. Our results provide a contrast to prominent media narratives of a large OB/GYN exodus from Texas. We contribute to a rapidly emerging literature by providing evidence that private enforcement had no detectable effect on the Texas physician labor market above and beyond the broader abortion policy environment.

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1 Introduction

Texas Senate Bill 8, enacted in September 2021, implemented a ban on abortions after detection of a fetal heartbeat (around six weeks of gestation) using a novel private enforcement mechanism to circumvent judicial review. Unlike all other abortion restrictions enforced by the state, SB-8 authorized “any person” to bring private lawsuits against anyone who “aids or abets” an abortion. The so-called “bounty law” could have been challenged in the courts, but the private enforcement mechanism was included for the express purpose of helping Senate Bill 8 survive judicial review since no state official was directly involved in enforcement.

Private enforcement potentially imposes different deterrent effect and risk compared to state enforcement because of two key differences. First, state enforcement requires a burden of proof that is “beyond a reasonable doubt,” but private enforcement only requires a “preponderance of the evidence.”¹ Second, the universe of potential plaintiffs is vastly larger than the universe of prosecutors. Imagine the less politically charged example of shoplifting. Under traditional state enforcement, a store would accuse a customer of shoplifting; the police would gather evidence; the prosecutor will review the evidence and may decide to make an arrest; the customer would be arraigned before a grand jury which decides if there is enough evidence to continue with prosecution; finally, the customer goes to trial to determine their guilt and punishment, if any. Under private enforcement, anyone could accuse a person of “aiding or abetting” shoplifting by filing a lawsuit for a minimum of \$10,000 in damages. This process skips the police investigation, prosecutor, arraignment, and grand jury. Unlike other private litigation, SB-8 also prohibits the accused from recovering their legal fees if they prevail. Both state and private enforcement impose a cost on behavior, but the differences are in the burden of proof and who may bring a claim.

Physicians weigh the expected benefit against the expected cost of practicing in Texas,

¹A preponderance of evidence means there is more than a fifty percent chance that the party being sued is at fault. Proof beyond a reasonable doubt means that a reasonable person would have no doubt about the defendant’s guilt.

and an increase in the expected cost could cause physicians to leave Texas. Expected cost in this context include the penalty for violating Texas’s abortion law multiplied by the probability of detection and conviction. Under state enforcement of abortion restrictions, that expected cost would fall exclusively on the physician who performs the procedure. Physicians who do not directly perform abortions would not consider that expected cost as part of their location decision. Under SB-8’s private enforcement, the expected cost changes in two important ways. First, the “preponderance of the evidence” standard raises the probability of liability relative to state-enforced prosecution. Second, the “aid or abet” language extends potential liability from exclusively abortion providers to any physician who manages pregnancy complications, counsels patients about their options, or provides emergency obstetric care. The result is an increase in expected cost that applies even to physicians who never perform an abortion. This framework predicts that the structure of SB-8 increased expected operating costs for reproductive health physicians (especially OB/GYNs, family medicine physicians, and emergency medicine physicians) which generated direct financial incentives to exit Texas. These incentives may also spillover onto other specialties, including oncology where treatment may harm a fetus. Our framework predicts that the supply of reproductive health physicians in Texas will fall leading to an increase in compensation for reproductive health physicians and an increase in job postings for reproductive health physicians. Prominent media narratives reported the SB-8 period as an exodus of OB/GYNs from Texas ([Petterson and Eubank, 2024](#); [Taladrid, 2024](#)), but the fixed costs of interstate practice relocation are high: approximately \$150,000 to \$250,000.

This paper examines migration responses to private enforcement mechanisms in the context of reproductive health physicians responding to Texas Senate Bill 8. SB-8 was introduced in March 2021, signed into law in May 2021, and implemented in September 2021 providing a natural experiment with one treated state. However, our analysis is unable to draw long-term conclusions from the effects of SB-8 because the U.S. Supreme Court overturned the constitutional right to an abortion in *Dobbs v. Jackson Women’s Health*

Organization nine months after SB-8 was enacted.

We use two complementary data sources to assess the change in the supply of reproductive health physicians and the consequences for compensation and job postings. We first use Medicare administrative data on physician locations from April 2015 through April 2022 (the last full quarter before the *Dobbs* ruling), covering 1,639,519 physicians, to a synthetic difference-in-differences (SDID) model ([Arkhangelsky et al., 2021](#)) comparing Texas to other states, before and after SB-8’s signing on May 19, 2021.² The SDID estimator reweights the donor states and pre-period quarters so that synthetic Texas tracks actual Texas’s own pre-period trajectory. Physician counts are normalized per 100,000 women of childbearing age. We first estimate triple-difference models comparing reproductive health physicians (OB/GYNs, family medicine physicians, and emergency medicine physicians) to all other physicians because they motivate the turn to SDID. Second, we use Wagescape data on physician job postings covering 1,327,394 postings from 2016–2022, including 107,474 OB/GYN and 150,839 family medicine postings. The posted salary is available for approximately 46 percent of postings. These data allow us to examine whether employer demand and compensation for reproductive health physicians in Texas shifted following SB-8.

We find no statistically significant migration response for reproductive health physicians in the thirteen months following SB-8’s signing. The Wagescape job posting data reinforce this null: both posted salaries and job posting volume for Texas reproductive health providers show no differential change relative to controls. The null results across both data sources and both identification strategies are consistent with the high fixed costs of interstate practice relocation. These findings are inconsistent with prominent media narratives characterizing the post-SB-8 period as an exodus of OB/GYNs from Texas ([Petterson and Eubank, 2024](#); [Taladrid, 2024](#)); our results do not support the conclusion that a large-scale departure occurred, at least prior to the *Dobbs* ruling.

²We measure treatment from SB-8’s signing rather than its September 1, 2021 enactment date because Google Trends data on searches for “Texas Heartbeat” show a large spike in attention at the time of signing. This response suggests that Senate Bill 8 was highly salient at its signing.

A substantial literature documents that state-enforced abortion restrictions reduce the supply of reproductive health physicians, with effect sizes scaling in the severity of enforcement. In the TRAP era predating *Dobbs*, [Chatterji et al. \(2025\)](#) find a 6.6 percent reduction in OB/GYN supply following targeted abortion provider regulations using a stacked difference-in-differences design, with evidence pointing to workforce exit rather than geographic relocation. [Markowski and Vandenbroeck \(2025\)](#) estimate a 4.67 percent reduction in OB/GYN density over 1993–2021 under TRAP laws, with effects persisting for a decade in nonmetropolitan counties. The effects of state-enforced restrictions extend beyond the current workforce to the training pipeline: [Hammoud et al. \(2024\)](#) document a significant decline in residency applications to OB/GYN programs in abortion-restrictive states following *Dobbs*.

In the post-*Dobbs* era, state-enforced abortion bans have produced larger and faster responses. [Rader et al. \(2025\)](#) document a 35 percent loss of practicing OB/GYNs in Idaho within 27 months of a near-total state-enforced ban with felony charges. [Zhu et al. \(2025\)](#) find a 4.2 percent decrease in OB/GYN practitioners per 100,000 reproductive-aged women in the most restrictive states, and [Diaz-Campo and Pineda-Torres \(2026\)](#) find a 3 percent decline across ban states. Across settings, effect sizes appear to scale with the severity and certainty of state enforcement, suggesting it is an important driver of the exit margin. Our design’s minimum detectable effect (4.1% at 80 percent power; Section 5) is smaller than the effect sizes reported by [Zhu et al. \(2025\)](#), [Markowski and Vandenbroeck \(2025\)](#), and, by a wide margin, [Rader et al. \(2025\)](#), meaning our design would likely have detected an effect of that magnitude had one occurred; it is larger than the estimate in [Diaz-Campo and Pineda-Torres \(2026\)](#), so we cannot rule out that an effect as small as theirs went undetected here.

Against this backdrop, several studies find null or modest migration effects for established physicians in the short run. [Staiger et al. \(2025\)](#) study practice locations for 60,085 OBGYNs across all policy environments from January 2018 to September 2024 and find no

statistically significant change in counts across banned, threatened, and protected states, with approximately 95 percent remaining in their pre-*Dobbs* policy environment. [Strasser et al. \(2024\)](#) similarly find no significant post-*Dobbs* shift in new-state enrollment flows comparing ban and no-ban states. [Nelson and Witko \(2026\)](#) document a preference-behavior gap: physicians in survey experiments express a preference for non-ban states, but actual migration effects remain modest. This behavioral stickiness is consistent with high fixed costs of relocation: [Falcettoni \(2021\)](#) identifies strong location preferences near residency training as a primary driver of physician location choice, and [Dench et al. \(2025\)](#) find significant out-migration only from states with total state-enforced bans, not from states with partial restrictions. [Bergmann \(2026\)](#) similarly finds no reduction in the aggregate post-*Dobbs* supply of OB/GYNs in ban states, but documents that the workforce’s political and demographic composition shifted sharply, with Democratic and female OB/GYNs becoming less likely to remain; this decomposition of an aggregate null into a compositional or destination-level margin motivates our own extensive/intensive-margin approach to the migration outcomes in Section 5. No prior work has examined whether private enforcement, which broadens the set of physicians facing expected operating costs, produces a different migration equilibrium.

2 Background: SB-8 and Private Enforcement

Texas Senate Bill 8 (SB-8) was signed on May 19, 2021 and enacted on September 1, 2021. SB-8 prohibits abortion after the detection of embryonic cardiac activity which typically occurs around six weeks of gestation. The statute’s enactment predated the Supreme Court’s decision in *Dobbs v. Jackson Women’s Health Organization* by nine months, placing it within the constitutional framework established by *Roe v. Wade* and *Planned Parenthood v. Casey*. Under precedent, SB-8 could have been ruled unconstitutional. The statute survived pre-enforcement judicial review through an unprecedented enforcement mechanism

that delegated enforcement exclusively to private litigation rather than state officials. Chief Justice John Roberts characterized this approach as “not only unusual, but unprecedented,” warning that “it is the role of the Supreme Court in our constitutional system that is at stake.” SB-8 avoided constitutional challenge by eliminating any state official against whom plaintiffs could seek injunctive relief.

The timing and scope of SB-8 merit particular attention for understanding its impact on physician practice patterns. At the time of enactment, approximately 85-90% of abortions performed in Texas occurred after six weeks of gestation, meaning the statute effectively prohibited the vast majority of abortion procedures previously performed in the state. Unlike subsequent abortion restrictions that took effect following *Dobbs*, SB-8 operated in a legal environment where abortion remained constitutionally protected nationally, creating a localized regulatory shock specific to Texas providers and patients.

SB-8 authorizes “any person, other than an officer or employee of a state or local governmental entity,” to bring private action against anyone who performs or “aids or abets” an abortion prohibited under the statute. This “aid or abet” language extends potential liability beyond physicians performing procedures to encompass a broad range of supportive activities, including providing transportation, financial assistance, counseling, or administrative support for prohibited abortions.

The incentives embedded in private enforcement substantially favor plaintiffs. Successful litigants receive minimum statutory damages of \$10,000 per abortion plus injunctive relief and attorney’s fees. Defendants cannot recover attorney’s fees or costs even if they win in litigation, creating asymmetric exposure to legal expenses. This one-way cost structure distinguishes private enforcement from typical private litigation, where prevailing defendants may recover costs, and from state prosecution, where defendants face no financial penalty for successful defense beyond legal fees. The statute further advantages plaintiffs by eliminating the heightened burden of proof required in state enforcement proceedings; private litigation requires only a preponderance of evidence rather than proof beyond reasonable doubt.

A single prohibited abortion could generate multiple lawsuits from different plaintiffs, each seeking the \$10,000 statutory minimum. For potential defendants, the expected cost of private enforcement exposure reflects not only the probability of liability but also the probability of facing litigation itself, multiplied by defense costs that cannot be recovered. Standard medical malpractice insurance covers claims that a physician’s negligence harmed a patient during care. SB-8 claims are different: they allege that a physician aided or abetted someone else’s abortion, not that the physician was negligent toward a patient. Because SB-8 claims fall outside what malpractice insurance is designed to cover, physicians face SB-8 defense costs with no possibility of fee recovery and no insurance coverage (Ivey, 2021).

Despite the statute’s theoretical enforcement potential, actual litigation activity under SB-8 was limited. Only three lawsuits were filed pursuant to the statute, all between September 20–22, 2021, and all targeting a single physician: Dr. Alan Braid of San Antonio. Dr. Braid had deliberately invited litigation by publishing an op-ed in *The Washington Post* announcing that he had performed an abortion in violation of SB-8. No additional lawsuits have been filed under SB-8’s provisions, no plaintiff has collected the statutory bounty, and no case has resulted in a finding of liability. The sole case to reach judicial resolution, *Gomez v. Braid*, was dismissed for lack of standing in December 2022 and affirmed on appeal in February 2024, establishing that the statute’s “any person” language remains subject to Texas constitutional standing requirements.

The absence of broader enforcement activity reflects strategic inaction by abortion opponents rather than lack of interest or resources. Major anti-abortion organizations, including Texas Right to Life and Operation Rescue, explicitly declined to file SB-8 lawsuits to avoid the creation of test cases that could provide grounds for invalidating the statute. By maintaining the threat of enforcement while limiting actual litigation, these organizations preserved SB-8’s deterrent effect while minimizing legal risk to the statute. Abortion providers in Texas ceased performing procedures after detection of embryonic cardiac activity upon SB-8’s effective date, consistent with deterrence operating through the threat of litigation

rather than its enforcement.

The persistent threat of litigation inherent in private enforcement, whereby any patient, family member, or stranger may initiate suit, differs from state enforcement risk in a key way. State prosecution requires a willing prosecutor; private enforcement requires only a willing plaintiff for whom the statute creates affirmative financial incentives and eliminates downside risk. For physicians evaluating practice location, private enforcement exposure represents an uninsurable increase in expected operating costs.

3 Conceptual Framework

Becker’s (1968) model of rational deterrence treats the decision to engage in a regulated activity as a function of expected liability costs: the probability of detection multiplied by the penalty conditional on detection, weighed against the expected benefit of the activity. SB-8 is designed to alter both components of this expected cost in ways that are qualitatively distinct from traditional state enforcement.

Probability of detection. State enforcement targets the physician who performs the procedure, requiring that a state actor identify, investigate, and prosecute a specific individual. SB-8’s “aid or abet” language dramatically broadens the set of actors subject to potential liability. A physician who provides standard obstetric care or counseling to a patient who subsequently obtains an abortion may be named in a private suit without having performed or directly facilitated an abortion. The vagueness of the statute’s scope means that even a physician taking no deliberate part in an abortion faces a positive and difficult-to-quantify probability of being drawn into litigation. This ambiguity itself constitutes a cost: physicians cannot insure against risks they cannot precisely estimate.

Penalty structure. Under state enforcement, a defendant who is acquitted at trial pays no penalty to the state and does not have to reimburse the prosecution’s costs. SB-8 works differently: even a physician who wins an SB-8 lawsuit cannot recover the legal fees spent

defending it, meaning a successful defense still imposes a net financial cost. Moreover, a single prohibited procedure can generate multiple independent lawsuits from different plaintiffs, each seeking the \$10,000 statutory minimum, creating effectively unbounded aggregate exposure from a single clinical event. Malpractice insurance offers no help under either system. It generally excludes state-enforced charges, and because it only covers negligence-based claims, it also does not cover SB-8’s aiding-or-abetting claims ([Business Insurance, 2023](#); [Ivey, 2021](#)).

Together, the broader detection margin and the asymmetric penalty structure imply that SB-8 imposes expected costs on a much wider set of physicians than traditional state enforcement, including physicians who provide no abortion services but whose routine clinical activities might be characterized as “aiding” a prohibited abortion. The predicted behavioral response is either to relocate away from Texas (the migration margin) or to demand higher compensation for bearing this incremental risk (the wage margin). The evidence in this paper finds no detectable response on either margin, consistent with the high fixed costs of interstate practice relocation estimated at \$150,000–\$250,000.³

4 Data

4.1 Sample Group: Regulated and Less Regulated States

Texas was not the only state facing a restrictive abortion environment during the analysis period. Using all states as the control group would contaminate the comparison: physicians in states with open abortion laws are operating in a fundamentally different policy atmosphere compared to a physician in a state that restricts abortion. We take [Dench et al.](#)

³This range reflects the main direct cost components documented in industry sources: malpractice tail coverage of 150–300 percent of the physician’s final annual premium (2021 Medical Liability Monitor Annual Rate Survey; Diederich Healthcare), lost billable revenue during the 90–150 day hospital credentialing and insurance-panel enrollment process (National Association Medical Staff Services benchmarks; Neolytix), real estate transaction costs, and state licensing and DEA registration fees. The American Medical Association estimates total physician turnover and replacement costs of \$500,000 to \$1,000,000 per physician.

(2025)’s definition of “hostile” and “total ban” states and bin them together as *regulated* states ($n = 25$); we take all other states ($n = 25$) and bin them together as *less regulated* states. The regulated states’ policy environment most closely resembles Texas’s policy environment before and around SB-8.⁴

This restriction serves a substantive identification purpose beyond improving pre-period balance. Physicians practicing in regulated states already operate under a restrictive abortion environment. A comparison of Texas RHPs against RHPs in these states therefore isolates the *marginal* effect of SB-8’s private enforcement mechanism above and beyond the effect of a restrictive abortion policy environment. By contrast, a comparison against physicians in less regulated states would conflate the private enforcement-specific deterrent with the broader effects of practicing in any abortion-restrictive state.

Table 4 is shown after an explanation of the data and confirms that the regulated-state sample is more similar to Texas on pre-period physician market outcomes than the full pool of states.⁵

4.2 Migration Data

This paper uses Medicare administrative data to construct a quarterly panel tracking physician practice locations over time. The primary data sources are the National Plan and Provider Enumeration System (NPPES) and the Centers for Medicare and Medicaid Services (CMS) Doctor and Clinician (DAC) database. The NPPES serves as the authoritative registry of all healthcare providers in the United States who hold a National Provider Identifier (NPI), a unique identification number required for billing Medicare and private insurance. The DAC database supplements NPPES by providing detailed practice location information for physicians who bill Medicare. Because practice addresses must be current to receive Medicare reimbursement, these administrative records offer high-quality location

⁴The 25 regulated states are: AL, AR, AZ, FL, GA, IA, ID, IN, KY, LA, MO, MS, NC, ND, NE, OH, OK, PA, SC, SD, TN, UT, WI, WV, and WY.

⁵Appendix R extends the control group to all states as a robustness check.

data with strong incentives for accuracy and timeliness.

The sample spans April 2015 through April 2022, creating a quarterly panel that covers 1,639,519 physicians. This period begins with the earliest quarter available in the NPPES-based panel and concludes with the last full quarter before the Supreme Court’s *Dobbs* decision in June 2022.⁶ The panel includes both physicians already in practice and new physicians entering the workforce from residency programs, providing broad coverage of the physician labor market over the study period. For each physician, the data contain the provider’s name, NPI, practice street address, and specialty designation. When a physician holds multiple taxonomy codes, we exclude any code identifying the individual as a student in an organized health care education or training program before selecting a primary specialty; NPPES/DAC do not otherwise record training status, so residents practicing under a given specialty (e.g., OB/GYN) are coded identically to attendings in that specialty and cannot be separately identified or excluded from the sample.

Five dependent variables capture different dimensions of physician mobility. First, a physician is coded as “active in the state” if she appears in the data with a practice location in that state during a given quarter. Second, a physician is classified as having “exited the state” if she leaves for another state while remaining in active practice. Third, a physician “entered the state” from another state while remaining in active practice. Fourth, “new entrance” identifies each physician’s very first active quarter in the data, capturing a new graduate making their entry into the workforce. Fifth, “retirement” captures permanent career exit by identifying each physician’s final active quarter. Summary statistics are shown in Table 1.

⁶Appendix C extends the sample one additional quarter past *Dobbs* as a robustness check

Summary Statistics

	All	TX RHP	TX Non-RHP	Non-TX RHP	Non-TX Non-RHP	(Min, Max)
<i>Panel A: NPI Physician Panel</i>						
Active Physicians	946 (455)	410 (21.24)	1176 (64.34)	540 (83.82)	1364 (263)	(354, 2246)
Exit State	4.78 (4.34)	1.02 (0.470)	4.22 (2.02)	2.23 (1.56)	7.50 (4.71)	(0, 42.25)
Enter State	4.72 (4.22)	1.25 (0.586)	5.35 (2.55)	2.06 (1.43)	7.49 (4.41)	(0, 29.97)
New Entrances	4.63 (18.58)	2.42 (9.02)	5.97 (20.77)	3.11 (12.25)	6.18 (23.31)	(0, 259)
Retirements	3.32 (19.78)	1.73 (9.77)	4.06 (22.30)	2.30 (13.20)	4.39 (24.79)	(0, 296)
<i>Panel B: Wagescape Job Postings</i>						
Posted Salary	184,510 (116,651)	150,084 (104,744)	180,711 (106,084)	175,857 (115,855)	187,716 (118,228)	(10,266, 863,998)
Log Posted Salary	11.93 (0.689)	11.64 (0.818)	11.94 (0.633)	11.83 (0.772)	11.95 (0.671)	(9, 13.67)
Job Postings	46.13 (68.27)	8.82 (6.89)	37.65 (28.70)	19.37 (26.34)	74.69 (85.89)	(0.1, 927)

Table 1: Summary statistics for both data sources. Columns show means by group: all observations, Texas RHP, Texas non-RHP, non-Texas RHP, non-Texas non-RHP (regulated states), and overall (Min, Max). SD in parentheses below means. Panel A: NPI physician panel; unit is the state $\times$ quarter cell collapsed to RHP and non-RHP groups; all Panel A outcomes (Active Physicians, Exit State, Enter State, New Entrances, Retirements) are normalized per 100,000 women of childbearing age. Panel B: Wagescape job postings restricted to the analysis sample; salary unit is the individual posting and is BEA RPP cost-of-living-adjusted in USD; Job Postings is the state $\times$ quarter cell count, normalized per 100,000 women of childbearing age. Sources: National Plan and Provider Enumeration System (NPPES)/CMS Doctors and Clinicians file (DAC); Wagescape via Dewey Data Platform.

Physician specialties are grouped into *reproductive health providers* (RHPs) and all other physicians. RHPs include obstetricians and gynecologists (OB/GYNs), family medicine

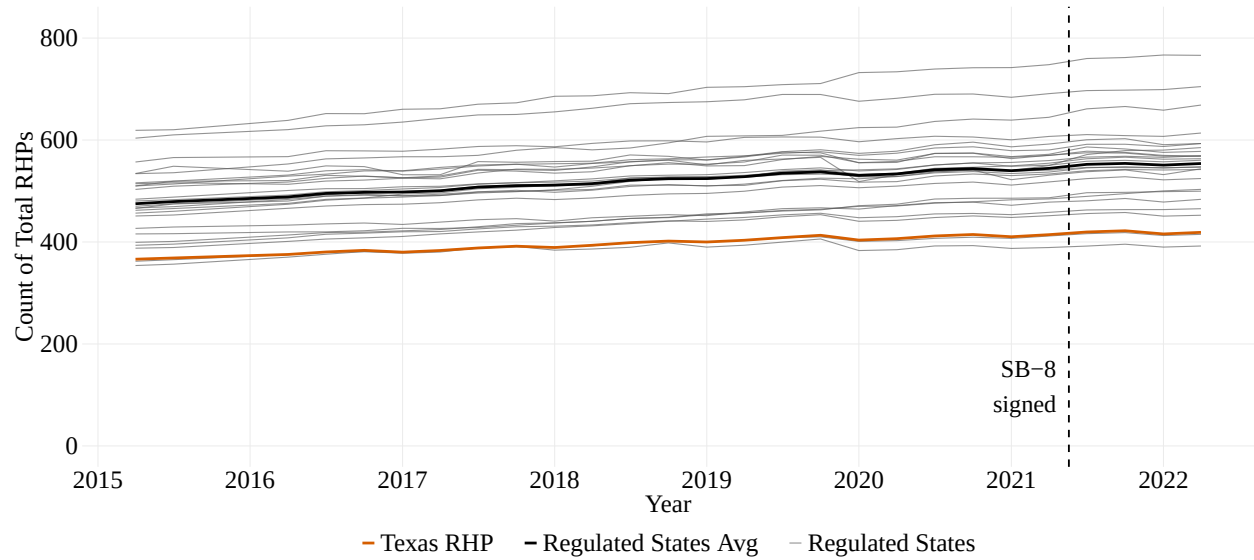
physicians, and emergency medicine physicians. OB/GYNs face the most direct exposure to SB-8’s provisions.⁷ Family medicine physicians are included because they may serve as the sole maternal healthcare provider in rural communities and encounter pregnancy complications as part of primary care. Emergency medicine physicians are included because they face time-sensitive decisions in obstetric emergencies where the line between permissible and impermissible treatment is ambiguous under the statute’s “aid or abet” language.⁸ All outcomes are normalized per 100,000 women of childbearing age in each state-quarter to account for the substantial variation in population across states.⁹

Figure 1 shows active reproductive health providers (RHPs) per 100,000 women of childbearing age and RHPs as a share of all active physicians. Each figure shows Texas (bold orange) alongside the average of the regulated states (bold black) and the trajectory of each individual regulated state (thin gray lines). The Texas series tracks the control average closely throughout the pre-period and shows no visible divergence following SB-8’s signing in May 2021.

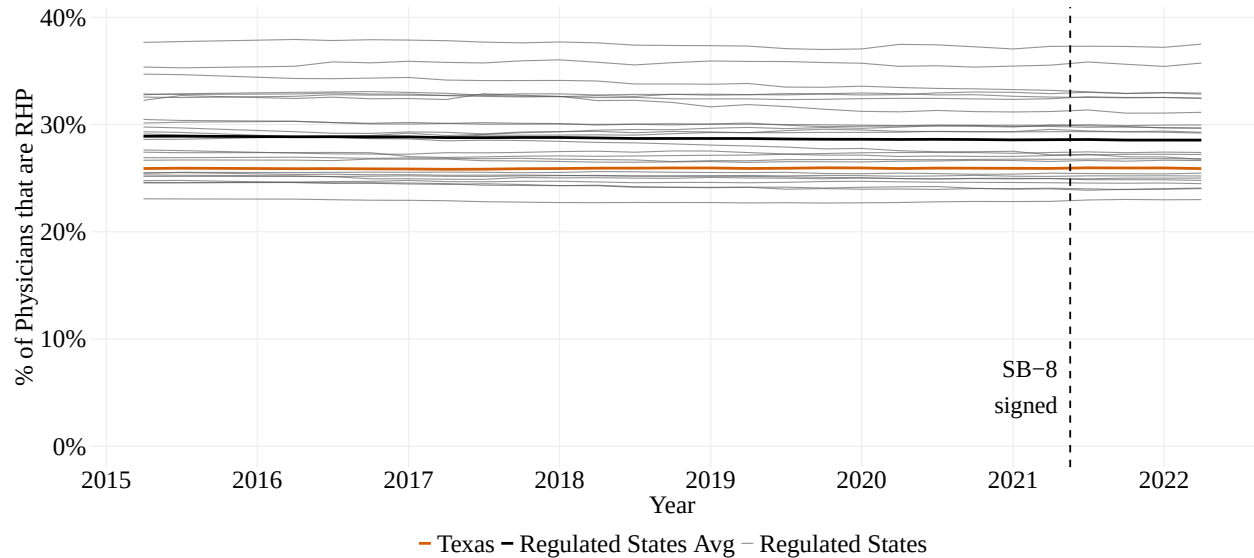
⁷Results from treating only OB/GYNs as treated can be found in Appendix B.

⁸Because of the ambiguity of SB-8’s reach, some specialties may be in a gray-zone of treatment. Appendix M removes gray-area specialties like general practice, pediatrics, and oncology from the control group.

⁹Women of childbearing age are defined as ages 15–44, following standard practice in the obstetrics literature.



(a) Active RHPs per 100k women: Texas vs. Regulated States.



(b) RHPs as a share of all active physicians (%): Texas vs. Regulated States.

Figure 1: Active reproductive health provider (RHP: OB/GYN, Family Medicine, and Emergency Medicine) counts per 100,000 women of childbearing age (ages 15–44) from 2007 to 2022. Panel (a) plots RHP counts per 100k women; panel (b) plots the RHP share of all active physicians. Both panels show Texas (bold orange), the average of the regulated states (bold black), and each individual regulated state (thin gray lines). The dashed vertical line marks May 19, 2021, the date SB-8 was signed. Source: NPPES/CMS DAC.

The universal coverage and longitudinal structure of these data make them well-suited for analyzing physician location decisions in response to state-level policy changes, and the administrative nature of the data reduces concerns about survey non-response or selective

attrition. The resulting data are a panel of specialty-state-quarter counts of migration, normalized per 100,000 women of childbearing age.

A limitation in our data is that we observe a lower bound on physician retirements. In general, most physicians do not deactivate their NPIs when they retire, so we substantially undercount true retirements. The retirement margin is particularly important for our analysis because physicians planning to retire in the near future may exit early rather than bear the legal risk of continued practice under SB-8, and the fixed cost of retiring a few years early is lower than the fixed cost of relocating for mid-career physicians. A further discussion of this point is included in the next steps at the end of this paper.

4.3 Supplementary Data: Physician Career Stage (CMS DAC)

We supplement the NPES-based migration panel with the CMS Doctors and Clinicians (DAC) file, a separate Medicare administrative dataset that additionally reports each physician’s medical school graduation year. We use graduation year to construct *age cohort*, a physician’s years of experience as of the SB-8 treatment date (May 19, 2021), binned into five-year intervals. Because DAC draws on Medicare Part B enrollment rather than NPES registration itself, DAC coverage is not universal: of the 1,593,892 physicians active in the NPES panel during a DAC-covered quarter (67 quarterly DAC snapshots, April 2017 through May 2025), 56.5 percent match to a DAC record in at least one of their active DAC-covered quarters.

Table 2 reports this physician-level match rate by state group and specialty; a physician is assigned to whichever state group and specialty they were in for the most DAC-covered quarters. Match rates are broadly similar across Texas (52.8 percent), the regulated states (56.2 percent), and the less regulated states (57.4 percent). By specialty, emergency medicine physicians match at the *highest* rate everywhere (59.8–70.1 percent).

NPPEs–DAC Match Rate by State Group and Specialty

	Texas		Regulated states		Less-Regulated states	
	Rate	N	Rate	N	Rate	N
Family Medicine	53.0%	17,923	55.7%	108,718	60.2%	103,803
OB/GYN	57.2%	6,115	59.1%	32,063	59.8%	39,295
Emergency Medicine	59.8%	6,592	65.2%	36,524	70.1%	42,202
Other Specialties	51.9%	86,498	55.4%	493,094	55.9%	621,065
All Specialties	52.8%	117,128	56.2%	670,399	57.4%	806,365

Table 2: **NPPEs–DAC match rate by state group and specialty.** Physician-level union match rate: each physician active during the 29 quarters with DAC coverage (2017 Q2–2025 Q2) counts once, regardless of how many of those quarters they were active in. Rate is the share of physicians who matched to a DAC record in at least one of their active DAC-covered quarters; N is the number of unique physicians in the cell (the match-rate denominator). A physician’s state group and specialty group are assigned by whichever cell they were in for the most DAC-covered quarters (their modal cell), since both can change over a career. Regulated and Less-Regulated states follow the classification in Section 4. Source: NPPEs and CMS DAC.

Because DAC matching is not random, we check whether physicians who ever match to DAC (at any point in their career) differ systematically from physicians who never match. Table 3 compares the two groups on specialty mix, geography, tenure, and migration behavior. Matched and unmatched physicians differ on nearly every observable. The starkest difference is in mobility: physicians who match to DAC have a state-to-state migration event at some point in the panel at a rate of 18.7 percent, compared with 3.9 percent for physicians who never match. This gap persists within every tenure decile (it is not simply an artifact of matched physicians having more active quarters and therefore more opportunities to migrate) and is not explained by unmatched physicians falling outside DAC’s 2017–2025 coverage window (only 1.1 percent of unmatched physicians have no active quarter overlapping that window at all). One group is statistically indistinguishable across the two samples: the share of physicians ever active in Texas (8.2 percent in both groups, $p = 0.180$).¹⁰

¹⁰Table 3’s match rate (911,370 of 1,639,519 physicians, 55.6 percent) is computed over the full NPPEs panel and is consistent with Table 2’s physician-level rate of 56.5 percent, which restricts to physicians active

Selection into DAC Matching

	(1)	(2)	(2–1)
	Matched to DAC	Unmatched	Diff
% OB/GYN	5.1 (22.0)	4.5 (20.8)	0.6 (0.000)
% Family Medicine	14.9 (35.6)	14.1 (34.8)	0.8 (0.000)
% Emergency Medicine	6.4 (24.6)	4.1 (19.8)	2.4 (0.000)
% Other Specialty	73.6 (44.1)	77.3 (41.9)	-3.7 (0.000)
% Ever in Texas	8.2 (27.4)	8.2 (27.5)	-0.1 (0.180)
% Ever in Regulated State	46.6 (49.9)	43.3 (49.6)	3.3 (0.000)
% Ever in Less-Regulated State	56.1 (49.6)	50.7 (50.0)	5.3 (0.000)
Quarters Active	35.33 (10.49)	31.00 (14.24)	4.33 (0.000)
% Ever Migrated	18.7 (39.0)	3.9 (19.4)	14.8 (0.000)
Observations (NPIs)	911,370	728,149	

Table 3: **Selection into DAC matching.** NPI-level comparison of physicians who ever match to a DAC record (Column 1) versus physicians who never match (Column 2), over the full NPPES panel. SD in parentheses below each mean; Column (2–1) reports the unmatched-minus-matched difference with a two-sided Welch t -test p -value in parentheses below. Active quarters counts each physician’s total number of active quarters in the NPPES panel. Ever migrated indicates at least one state-to-state migration event (entry into or exit from a state while remaining in active practice) at any point in the panel. Source: NPPES and CMS DAC.

This selection pattern is consistent with Medicare-enrolled physicians having a stronger during a DAC-covered quarter; the small remaining gap is attributable to physicians whose NPPES tenure falls entirely outside DAC’s 2017–2025 coverage window, who cannot match by construction.

administrative incentive to keep their NPES address current when they relocate. Because the matched DAC subsample is disproportionately drawn from more mobile physicians, we flag this selection pattern as a limitation for any future cohort-specific analysis built on the DAC match.

4.4 Wagescape Job Postings Data

We use physician job posting data from Wagescape to analyze the offered salary and job posting volume for reproductive health physicians. These data cover 1,327,394 physician job postings for the period 2016–2022, including 107,474 OB/GYN postings, 150,839 Family Medicine postings, and 49,068 Emergency Medicine postings.¹¹ Each posting records the position state, date posted, and, if disclosed by the employer, the offered salary. Employers disclose salary in approximately 46 percent of postings, with coverage rising from 6–13 percent in 2016–2017 to 54–69 percent in 2018–2022. After limiting to the 50 US states and the District of Columbia, the salary analysis sample includes all postings with a positive reported salary across all three physician specialties in the RHP group.

A potential threat to identification in the wage analysis is differential trends in salary disclosure rates across groups. If Texas RHP employers became more or less likely to disclose posted salaries around SB-8’s signing (relative to Texas non-RHPs or to control-state RHPs) the estimated wage effect could reflect a composition change in the salary-reporting sample rather than a true price signal. Panel (a) of Figure 2 plots the quarterly share of postings with a salary listed for four groups: Texas RHP, control-state RHP, Texas non-RHP, and control-state non-RHP. The four series track closely throughout the sample period, including around

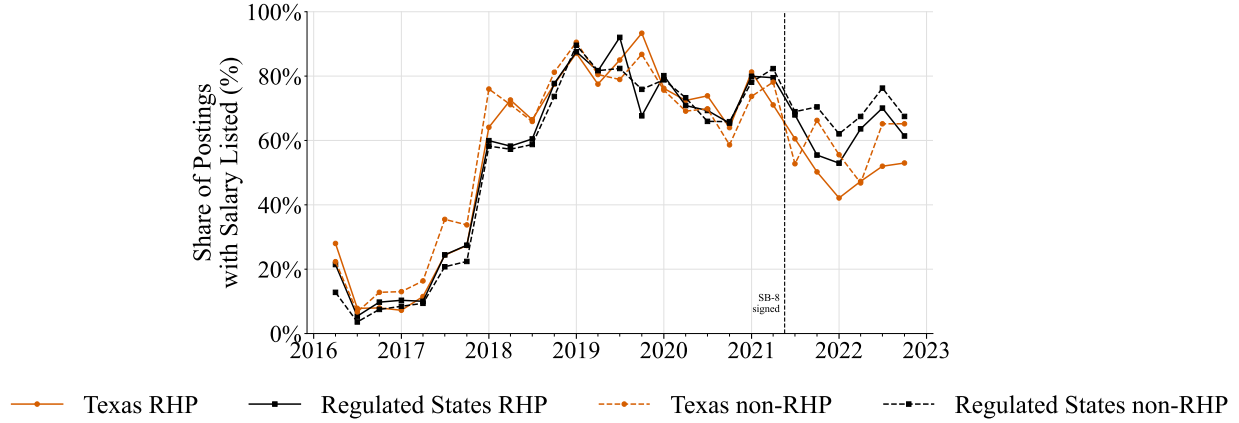
¹¹OB/GYN postings are identified using SOC code 29-1218. Family medicine postings are identified using SOC code 29-1215, and we remove Family medicine postings that include “no OB” in the job title. ER physicians are identified using the job title keywords “emergency medicine,” “emergency physician,” “ER physician,” “EM physician,” and “ED physician.” The NPI analysis classifies all Family Medicine physicians (taxonomy 207Q) as reproductive health providers, while the Wagescape wage analysis restricts to Family Medicine physicians whose job title indicates obstetric, emergency, or otherwise ambiguous reproductive content. In the NPI data, it is not possible to identify which family physicians perform obstetric services from the administrative records alone. This means the NPI RHP group is broader than the Wagescape RHP group.

the May 2021 signing date, with no visible differential break for Texas RHPs relative to any comparison group. This parallel behavior in disclosure rates supports the interpretation that changes in the salary sample composition are not a confounding factor in the wage estimates.

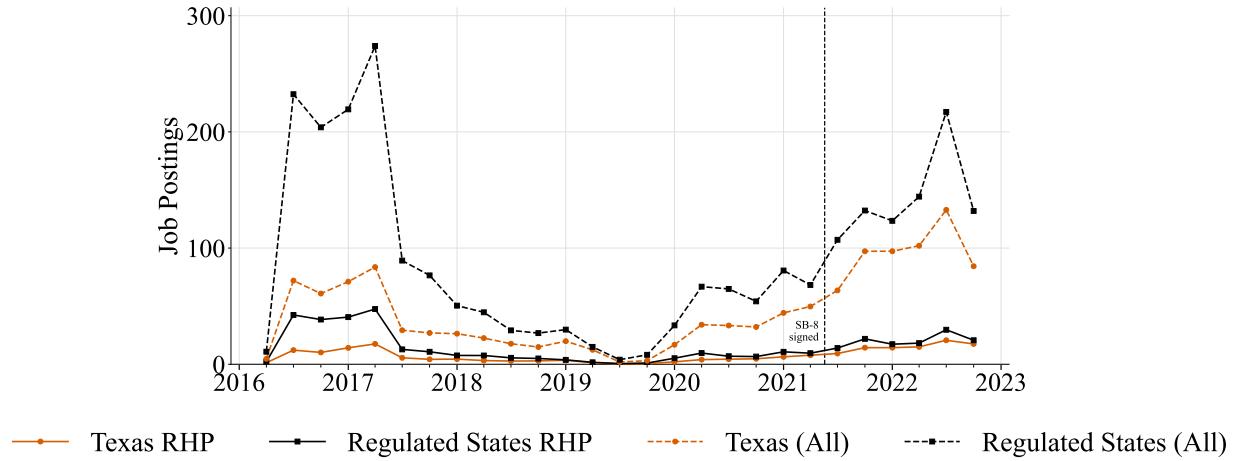
Panel (b) shows the total volume of physician job postings per 100,000 women of child-bearing age, comparing Texas to the control states for all physicians and for RHPs separately. Across all four series we observe substantial growth from 2016 through 2022, with particularly steep growth beginning in 2020. This growth trend is consistent across all groups and reflects Wagescape’s expanding coverage of the job market rather than an underlying increase in actual job openings. The parallel trends in the pre-SB-8 period and absence of a visible discontinuity at SB-8’s signing date support the identifying assumption. Texas’s OB/GYN share of total physician postings also rose over this period (approximately 5 to 13 percent, 2016–2022).¹²

Panel (c) shows posted salaries for RHP and non-RHP physicians in Texas and the control states. Because nominal salaries reflect both compensation and local cost of living, we adjust posted salaries using the Bureau of Economic Analysis’s Regional Price Parity (RPP) index, which measures the price level of goods and services in each state relative to the national average. We refer to this adjusted salary as “COL-adjusted” (cost-of-living-adjusted) throughout the paper. This adjustment matters for our comparison because Texas and the control states differ in cost of living: without it, a salary difference between Texas and the control states could reflect regional price levels rather than a true difference in physician compensation. Table 1 presents summary statistics for the regression sample.

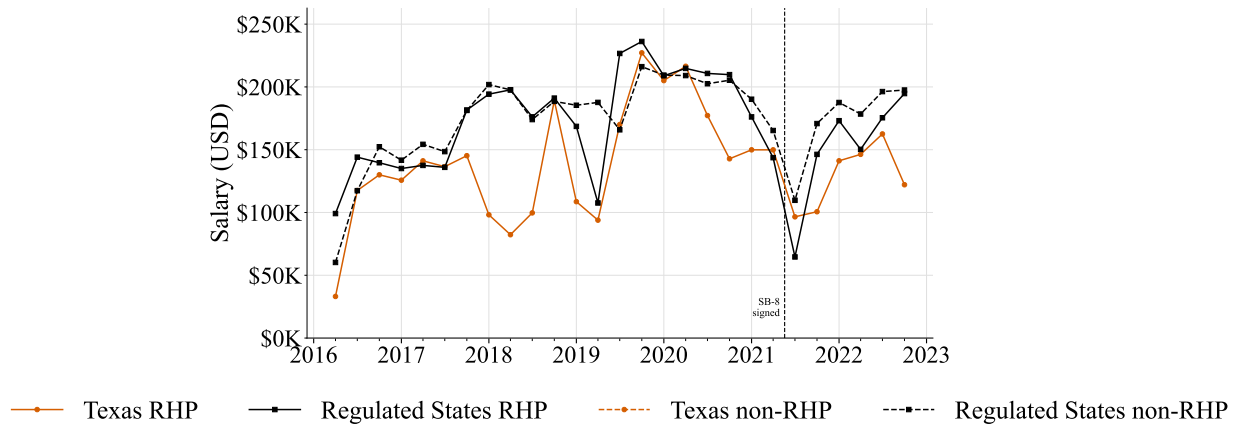
¹²Appendix G shows this growth is not distinctively Texan: Texas ranks 16th of 51 states by OB/GYN posting-share growth, consistent with broad-based Wagescape coverage expansion rather than a policy-driven compositional shift.



(a) Salary disclosure rate: share of postings with salary listed.



(b) Total job postings per 100k women.



(c) Posted salary.

Figure 2: Wagescape descriptive figures, all comparing Texas (orange) to the regulated states (black), with solid lines for RHP and dashed lines for all physicians. Panel (a): quarterly share of postings with salary disclosed; the four series track in parallel with no differential break around SB-8's signing, supporting the salary-sample validity assumption. Panel (b): total physician job postings per 100,000 women aged 15–44; post-2020 growth reflects Wagescape's expanding coverage rather than actual hiring. Panel (c): posted salary (BEA RPP cost-of-living adjusted) for RHP and non-RHP physicians in Texas and the regulated states. The dashed vertical line in all panels marks SB-8's signing date (May 19, 2021). Source: Wagescape via Dewey Data Platform.

4.5 Balance: Texas vs. Regulated States vs. Less Regulated States

Table 4 presents pre-period (pre-May 2021) means for all NPI regression outcomes across three groups: Texas, the regulated states, and the less regulated states. The purpose of the balance check is to assess whether the regulated states are a more credible counterfactual for Texas physicians than the less regulated states. For purposes of assessing balance, we use per-taxonomy cells rather than RHP vs. non-RHP cells.

Texas is a large, growing state with a distinctive physician labor market. In the pre-period, Texas has lower counts of active RHPs per 100k women (131.42 per taxonomy group) than both the regulated states (171.78) and the less regulated states (174.67), reflecting Texas’s rapid population growth relative to its physician supply. Across all six balance covariates jointly, the regulated states are a closer overall match to Texas than the less regulated states are: the multivariate normalized difference $\hat{\Delta}^{mv}$ of Imbens and Rubin (2015) is 1.941 for the regulated-state comparison versus 2.124 for the less-regulated-state comparison (Table 4, bottom row). This further motivates using the regulated states as the main specification: these states faced similar abortion-policy environments to Texas and are a closer overall pre-period match, providing a more credible counterfactual for identification.¹³

¹³Full-sample (all-states) results are qualitatively unchanged and can be found in Appendix R.

Balance: Texas vs. Regulated and Less Regulated States					
	(1)	(2)	(3)	(4)	(5)
	Texas	Regulated	Less Regulated	(2–1)	(3–1)
Count of Total RHPs	131.42	171.78	174.67	40.36	43.25
		(24.53)	(34.33)	(0.000)	(0.000)
Count of Total Non-RHPs	34.17	41.82	53.71	7.65	19.54
		(6.59)	(14.65)	(0.000)	(0.000)
RHP Salary (USD)	155087	175164	163593	20078	8507
		(14407)	(17671)	(0.000)	(0.024)
Non-RHP Salary (USD)	163471	179526	169724	16055	6254
		(14806)	(11202)	(0.000)	(0.010)
RHP Job Postings	5.91	17.74	21.51	11.83	15.60
		(10.72)	(14.33)	(0.000)	(0.000)
Non-RHP Job Postings	25.11	60.69	67.20	35.58	42.09
		(39.06)	(32.89)	(0.000)	(0.000)
Multivariate Distance ($\hat{\Delta}^{mv}$)				1.941	2.124
Observations (states)	1	25	25		

Table 4: **Balance: Texas compared to regulated and less regulated states.** Pre-period (before May 19, 2021), state-level means for Texas, the regulated states (25 states), and the less regulated states (25 states); each mean is followed on the row below by, where applicable, the across-state SD of state-level means within that group (Texas has no SD, as there is only one treated state) and, for the (2–1)/(3–1) columns, a one-sample t -test p -value. Columns (2–1) and (3–1) report the regulated- and less-regulated-state means minus the Texas mean. RHP salary is BEA RPP cost-of-living-adjusted; RHP counts and job postings are normalized per 100,000 women of childbearing age (15–44). The final (bolded) row reports the multivariate normalized difference $\hat{\Delta}^{mv}$ of [Imbens and Rubin \(2015\)](#) (Sec. 14.3, Eq. 14.9): the Mahalanobis distance between the Texas and comparison-group covariate means across all six covariates jointly, adapted for a single treated state by pooling the covariance matrix across all 50 non-Texas states; a smaller $\hat{\Delta}^{mv}$ indicates a closer match to Texas. Each state contributes a single pre-period mean to this table (collapsed across pre-period quarters before the across-state SD/ p -value is computed), so this table checks pre-period *level* balance across states; it does not test for differential pre-period *trends*, which the event-study pre-period coefficients in Section 5 address directly. Source: NPPES/CMS DAC and Wagescape.

4.6 Descriptive Trends: Difference-in-Differences

DiD: Treated physicians. Figure 3 compares Texas RHPs to control-state RHPs in a DiD event study. Texas entered the period with fewer active RHPs per 100,000 women of childbearing age than the control-state average, and that gap widens over the pre-period: pre-period coefficients slope downward, indicating divergence rather than convergence.¹⁴ A downward-sloping pre-period is inconsistent with the parallel trends assumption for a naive DiD of Texas RHPs against control-state RHPs: the divergence trend would continue absent any policy change, rendering a naive DiD estimate biased.

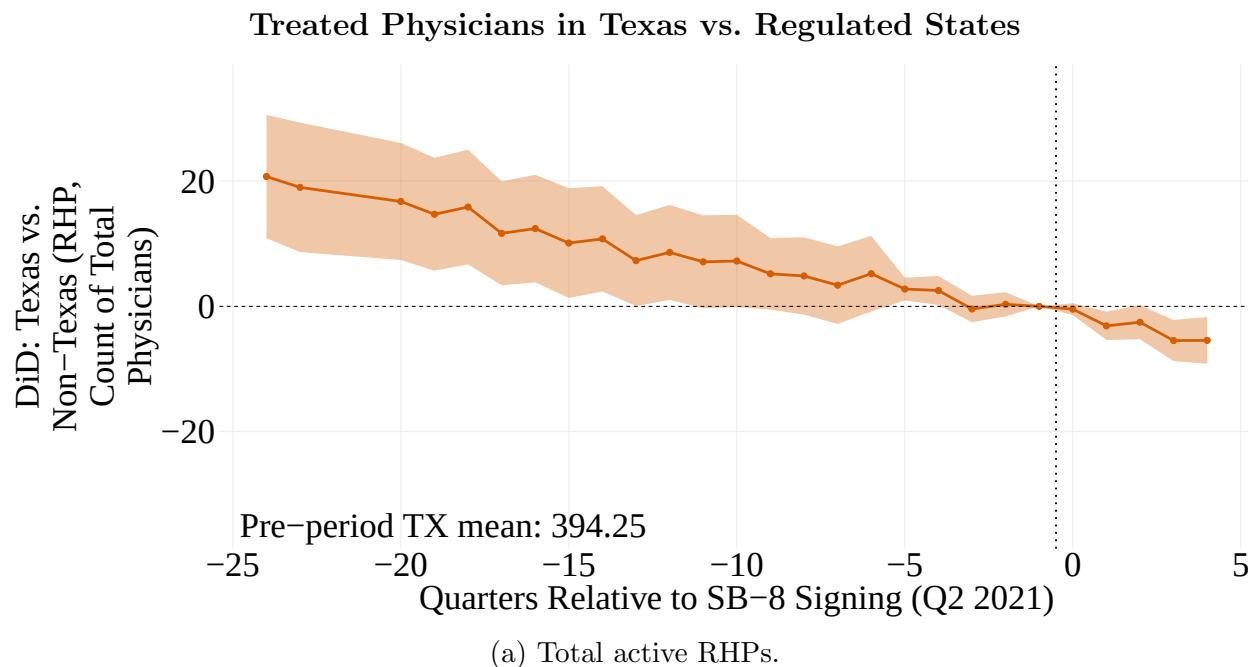


Figure 3: Difference-in-differences event study estimate for reproductive health providers (RHP: OB/GYN, Family Medicine, and Emergency Medicine), comparing Texas RHP physicians to non-Texas RHP physicians over time. The estimating equation includes state and date fixed effects; the reference period is Q1 2021 (event time -1). Each coefficient represents the quarterly Texas–non-Texas differential for RHP physicians relative to the reference quarter, per 100,000 women of childbearing age. Sample: 2015–2022 quarterly, regulated states; 95% confidence intervals shown. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

¹⁴This divergence is a denominator artifact. Texas physician counts grow in absolute terms over the sample period, but the female population aged 15–44 grows faster, so the per-100k stock declines even as the underlying physician count rises. Running the same event study on raw physician counts reverses the pre-period slope, confirming that the trend reflects population dynamics rather than physician behavior.

DiD: Control physicians. Figure 4 presents the same DiD for non-RHP physicians, those with no professional exposure to SB-8’s “aid or abet” provisions. The pre-period pattern closely mirrors Figure 3: Texas non-RHPs also exhibit downward pre-period divergence at a similar rate. Because the pattern appears in both the specialty group exposed to the private enforcement mechanism and the specialty group with no exposure, it cannot be attributed to SB-8. It reflects a general Texas physician-market trend. Attributing this trend to treatment would produce a biased estimate of the private enforcement effect.

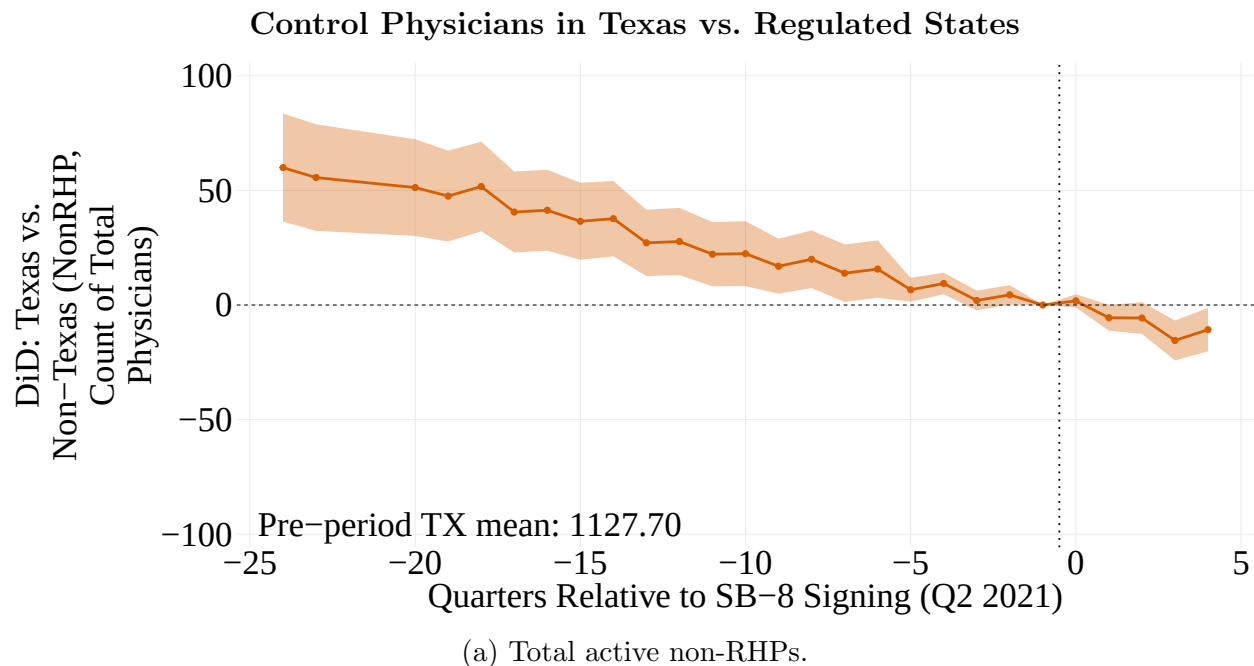


Figure 4: Difference-in-differences event study estimate for non-reproductive-health physicians (the control group), comparing Texas non-RHP physicians to non-Texas non-RHP physicians over time. The estimating equation includes state and date fixed effects; the reference period is Q1 2021 (event time -1). This figure serves as a placebo check: absent SB-8 exposure, the Texas differential should be flat throughout. Per 100,000 women of childbearing age. Sample: 2015–2022 quarterly, regulated states; 95% confidence intervals shown. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

5 Results

We report two identification strategies for every outcome. Our preferred estimates come from the synthetic difference-in-differences (SDID) estimator of [Arkhangelsky et al. \(2021\)](#), which reweights the donor (regulated) states and pre-period quarters so that synthetic Texas

matches actual Texas’s own pre-period trajectory, rather than assuming Texas and the control states would have moved in parallel absent treatment. Formally, following [Bergmann \(2026\)](#)’s adaptation of [Arkhangelsky et al.](#)’s estimator to a single-cohort setting, we estimate

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \sum_{s=1}^N \sum_t \left[\text{Outcome}_{st} - \mu - \underbrace{\alpha_s}_{\text{state FE}} - \underbrace{\beta_t}_{\text{quarter FE}} - \tau \cdot \underbrace{\text{Texas}_s \times \text{Post SB8}_t}_{\substack{=1 \text{ if Texas,} \\ \text{after May 19, 2021}}} \right]^2 \underbrace{\hat{\omega}_s}_{\substack{\text{unit} \\ \text{weight}}} \underbrace{\hat{\lambda}_t}_{\substack{\text{time} \\ \text{weight}}}$$

where

$$\text{Outcome}_{st} \in \{\text{Total Active, Leaver Share to Regulated, Entrant Share from Regulated, New Entrances, Retirements, Salary, and Job Posting Volume}\} \quad (1)$$

on the RHP-only panel (Texas as the single treated unit, the regulated states as the donor pool). The unit weights $\hat{\omega}_s$ are chosen so that the weighted average of the control states’ pre-period outcomes tracks Texas’s own pre-period trajectory as closely as possible; the time weights $\hat{\lambda}_t$ upweight the pre-period quarters that best predict the post-treatment path, following the same logic as synthetic control but applied to both dimensions jointly. $\hat{\tau}$ is the SDID estimator of interest: the differential change in outcomes for Texas relative to its synthetic counterfactual, following SB-8’s signing. Because the weights make the comparison local, emphasizing donor states and pre-period windows that reproduce Texas’s own trajectory, the identifying parallel-trends requirement is imposed on the weighted comparison rather than on the raw data, which is what lets SDID tolerate the residual pre-period trend that the DDD event studies below still show. Inference uses the placebo variance estimator with $B = 1,000$ replications, clustering implicitly through the leave-one-out donor-state resampling (Section [5.1](#)).

We also report the triple-difference (DDD) design as build-up: it motivates, empirically, why the parallel-trends assumption is worth relaxing, and it exploits the same three sources

of variation (time, geography, and specialty) using non-RHP physicians as an additional within-Texas control that SDID’s RHP-only panel does not use. Appendix A states the DDD estimating equation and its coefficient β_1 in full; we refer to β_1 throughout the Results and in Table A1 without repeating the equation here. As the DDD event studies below show, a residual pre-period slope is visible in several outcomes: this is the reason DDD is presented as build-up rather than as the paper’s preferred result. Appendix E additionally reports a synthetic control (SC) comparison (matching pre-period levels only, without time-weighting) as a robustness check.

Because Texas is the only treated state ($N_1 = 1$), conventional cluster-robust and wild cluster bootstrap standard errors are invalid for the scalar DDD coefficient (Ferman and Pinto, 2019). We use the heteroskedasticity-corrected closed-form standard error of Ferman and Pinto (2019) for DDD, which remains valid with a single treated unit. All migration outcomes are normalized per 100,000 women of childbearing age. Event studies (both DDD and SDID) use cluster-robust standard errors clustered by state, which remain valid when many time-varying coefficients are estimated simultaneously.

5.1 Results: Migration

We present migration results in three stages, each corresponding to one of the event study figures below. The comparison across figures motivates the DDD specification and illustrates what a naive DiD would miss. Table 5 presents the SDID, SC, and DDD scalar estimates for the three main-body migration outcomes (New Entrances and Retirements are reported in Appendix D’s full table instead); Appendix A reports the full DDD specification (Table A1).

SDID, SC, and DDD Estimates: Migration Outcomes

Outcome	SDID	SC	DDD
Count of Total RHPs	-2.877 (5.729)	0.793 (15.400)	23.020 (41.595)
Leaver Share to Regulated	0.049 (0.092)	0.011 (0.071)	0.032 (0.045)
Entrant Share from Regulated	0.045 (0.144)	0.054 (0.103)	0.049 (0.048)

Table 5: SDID, SC, and DDD estimates of the effect of SB-8 on the active RHP stock and the leaver/entrant destination-share outcomes (2015–2022 quarterly panel; New Entrances and Retirements are reported in Appendix D’s Table D1 instead). Count of Total RHPs is per 100,000 women of childbearing age; Leaver/Entrant Share are shares (0–1) of interstate movers landing in or arriving from a regulated state. SDID and SC operate on the RHP-only panel with Texas as the single treated unit; inference via placebo variance ($B = 1,000$). DDD uses [Ferman and Pinto \(2019\)](#) heteroskedasticity-corrected SEs, same specification as Appendix Table A1. Control group: regulated states (25 states). SEs in parentheses. Source: NPPES/CMS DAC.

Triple-difference estimates. The DDD design addresses this directly by differencing out the general Texas physician-market trend identified in Figure 4 (Section 4). Figure 5 shows the resulting DDD event study. The estimating equation nets out a nationwide RHP-versus-non-RHP trend and a Texas-wide trend common to both specialty groups as separate terms. A residual pre-period slope remains in the active count panel: this is precisely the parallel-trends violation that motivates treating SDID, not DDD, as this paper’s preferred migration estimate (Section 5.1 constructs a Texas counterfactual without imposing parallel

pre-trends).¹⁵

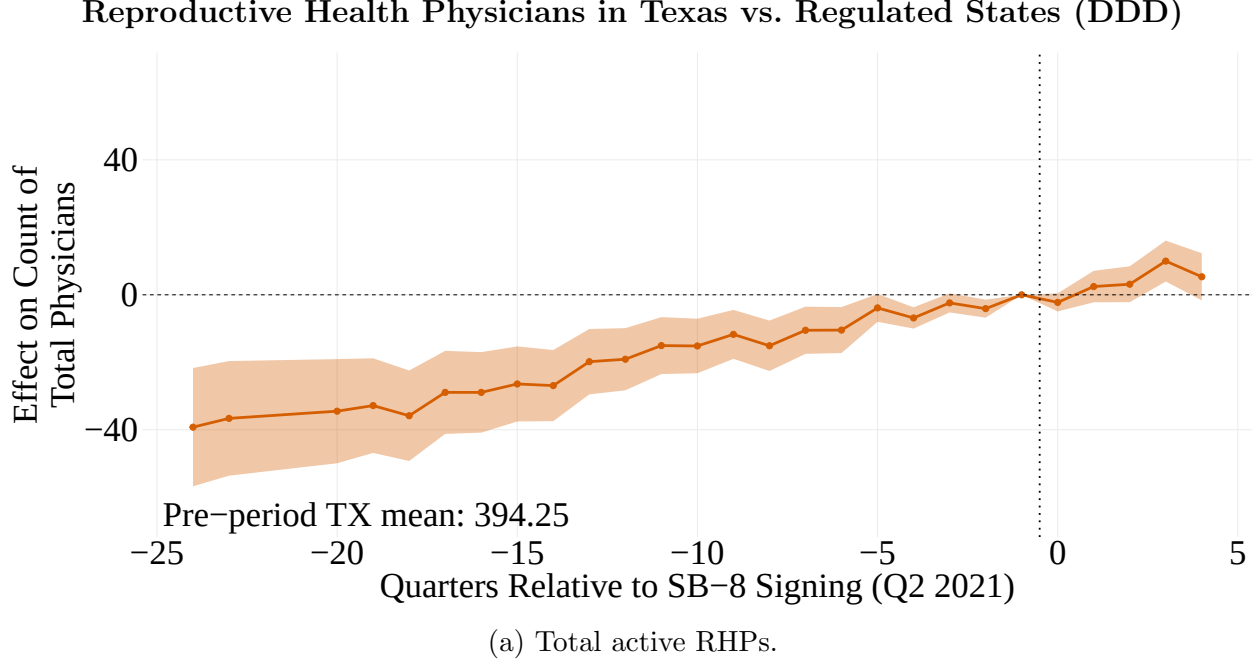


Figure 5: Triple-difference event study estimate for reproductive health provider (RHP: OB/GYN, Family Medicine, and Emergency Medicine) active physician stock. The estimating equation is $\text{Outcome}_{ist} = \alpha + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[\text{event_time} = k] \times \text{Texas}_s \times \text{RHP}_i + [\text{FEs, plus the all-Texas and all-RHP double-difference components, each allowed to vary by event time}] + \varepsilon_{ist}$. Coefficients β_k are quarterly treatment effects relative to Q1 2021 (reference period: event time -1), per 100,000 women of childbearing age. Unit (state $\times$ group) and date fixed effects are included. Sample: 2015–2022 quarterly, regulated states. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

This subsection presents the paper’s preferred migration estimates. The residual pre-period slope in the active physician stock event study (Figure 5) is an empirical motivation: it is a parallel-trends violation the DDD design cannot correct for, and it is why we do not treat the DDD β_1 estimates above as the paper’s preferred migration result. The Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021) constructs a reweighted donor pool of control states and a set of time weights jointly, so that the pre-period trajectory of synthetic Texas tracks both the level and trend of actual Texas, directly addressing the residual pre-trend that DDD leaves unresolved. It operates on the RHP-only panel, using Texas as the single treated unit and the regulated states as the donor pool. In-

¹⁵Appendix Table A1 presents the scalar DDD estimates in full.

ference uses the placebo variance estimator with $B = 1000$ replications. The same estimator is applied to the wage outcomes in Section 5.2.¹⁶

For the active RHP stock, the SDID estimate is $\hat{\tau} = -2.877$ per 100,000 women of childbearing age (placebo SE 5.729), and the SC estimate is directionally consistent (Table D1); both are statistically indistinguishable from zero, matching the DDD conclusion above without relying on the parallel-pre-trends assumption. This is a precisely estimated null, not merely an absence of evidence: at 80 percent power, the SDID standard error implies a minimum detectable effect of 16.1 per 100,000 women of childbearing age (4.1 percent of the pre-period Texas RHP baseline of 394.2, or approximately 989 physicians out of Texas’s 1,639,519-physician panel), and 18.6 per 100,000 (4.7 percent, $\approx 1,145$ physicians) at 90 percent power. Relative to the migration literature reviewed in Section 1, this design would likely have detected an effect the size of Zhu et al. (2025)’s 4.2 percent, Markowski and Vandenbroeck (2025)’s 4.67 percent, or Rader et al. (2025)’s 35 percent, but not necessarily an effect as small as Diaz-Campo and Pineda-Torres (2026)’s 3 percent. Appendix E reports the SC trajectory plots for all NPI outcomes.

Following Ciccia (2024)’s event-study extension of Clarke et al. (2024)’s SDID estimator, Figure 6 plots the per-quarter SDID treatment effect for each of the three main-body migration outcomes, with placebo-variance confidence intervals, rather than only the static comparison in Table D1. Because Texas is the only treated state and SB-8 has a single effective date, this collapses to the single-cohort case of Ciccia’s estimator: at each quarter the coefficient is the gap between Texas and synthetic Texas, net of the same time-weighted pre-period baseline that identifies the static SDID estimate in Table D1. Confidence intervals come from a leave-one-out placebo procedure: each regulated state is treated as a placebo-treated unit in turn, and the standard deviation of the resulting placebo effect paths at each quarter gives the reported interval. Pre-period coefficients are statistically indistinguishable from zero throughout, and no post-signing break is evident in any of the panels, consistent

¹⁶A tabular summary of all estimates (migration and wage), including a levels-only synthetic control (SC) comparison reported as a robustness check, is in Appendix D.

with the static-estimate results above.

SDID Event Study: Reproductive Health Physicians vs. Synthetic Texas

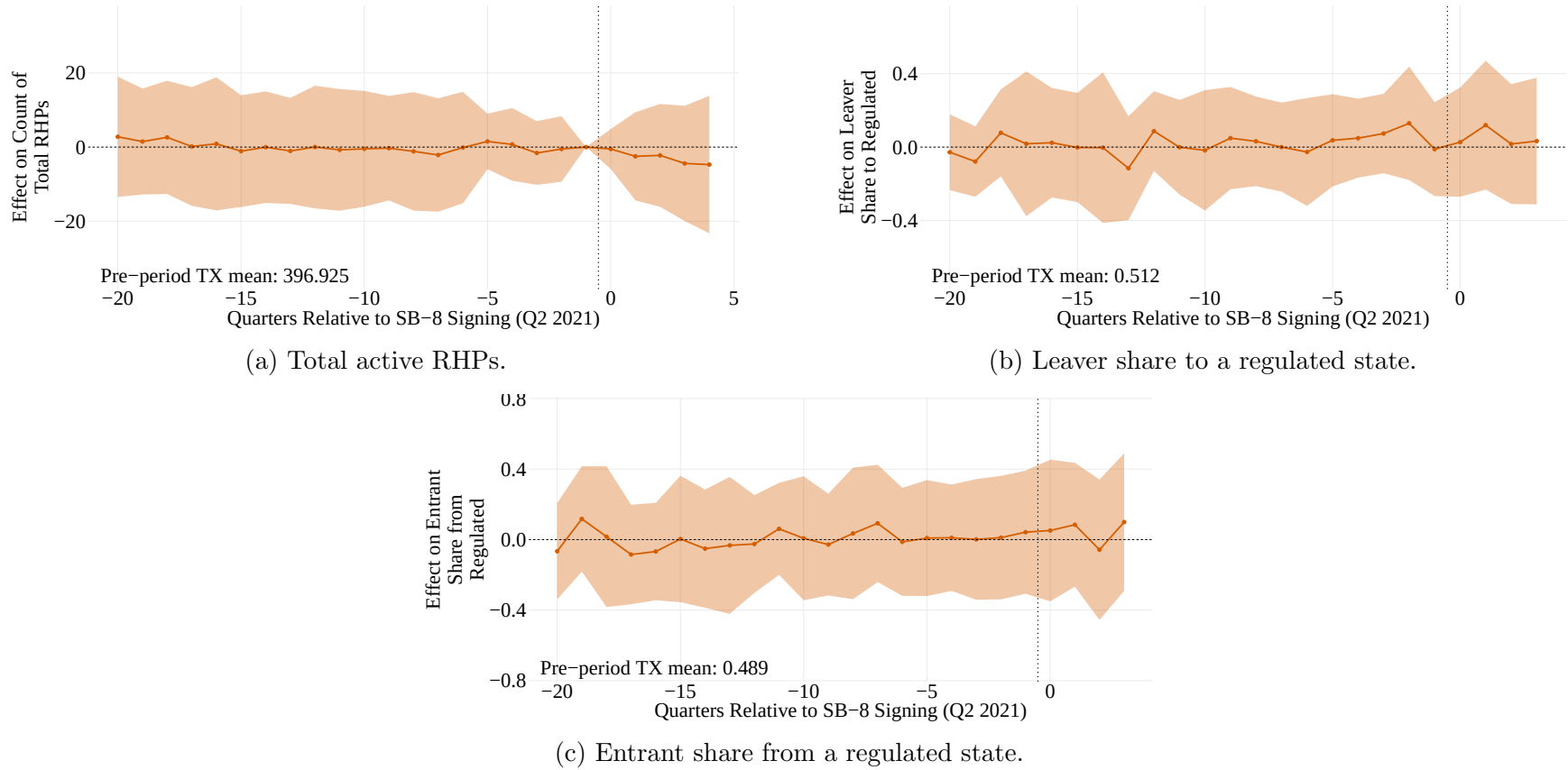


Figure 6: SDID event studies for three main-body migration outcomes, following [Ciccia \(2024\)](#). Each point is the per-quarter SDID treatment effect (Texas versus synthetic Texas, net of the time-weighted pre-period baseline); shaded regions show 95% confidence intervals from a leave-one-out placebo procedure (each of the 25 regulated states treated as pseudo-treated in turn). Quarters relative to SB-8 signing (May 19, 2021); event time 0 is Q2 2021. Total active RHPs is per 100,000 women of childbearing age; leaver/entrant share are shares (0–1) of interstate movers. Control group: regulated states. Source: NPPES NPI registry.

Figure 7 shows the SDID unit weights (ω) for one representative migration outcome and one representative wage outcome (Section 5.2).

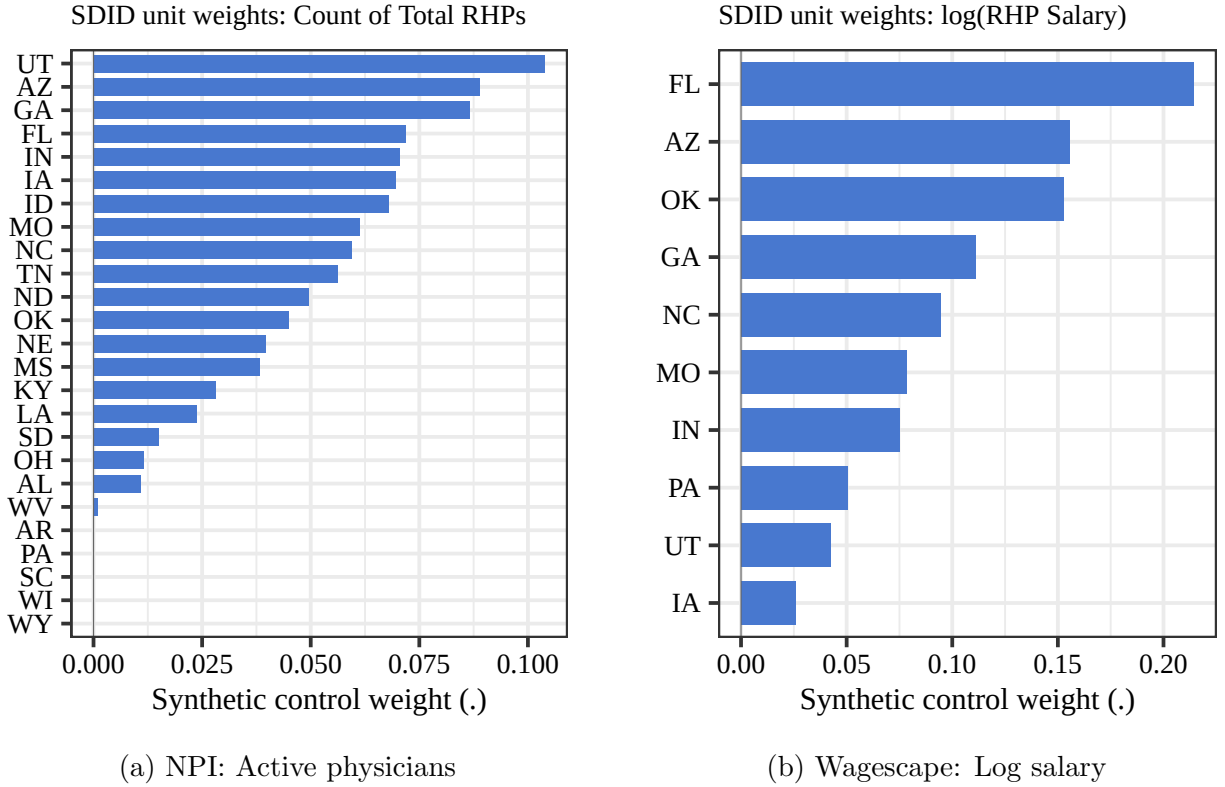


Figure 7: SDID unit weights (ω) for two representative outcomes. Each bar shows the weight assigned to a control state in constructing synthetic Texas. Control group: regulated states.

Figure 8 shows the SDID trajectory plots underlying the static estimates above: Texas (solid orange) against synthetic Texas (dashed black), for the same three main-body migration outcomes just shown as per-quarter event studies. SDID is designed to match the pre-period trend of Texas, and the synthetic counterfactual tracks the actual trajectory closely in each pre-period panel. No outcome shows a post-signing divergence between Texas and synthetic Texas. That the null result is consistent whether the synthetic counterfactual is calibrated to levels (SC, Appendix E) or to trends (SDID) indicates that the finding does not depend on the particular weighting scheme used to construct synthetic Texas.

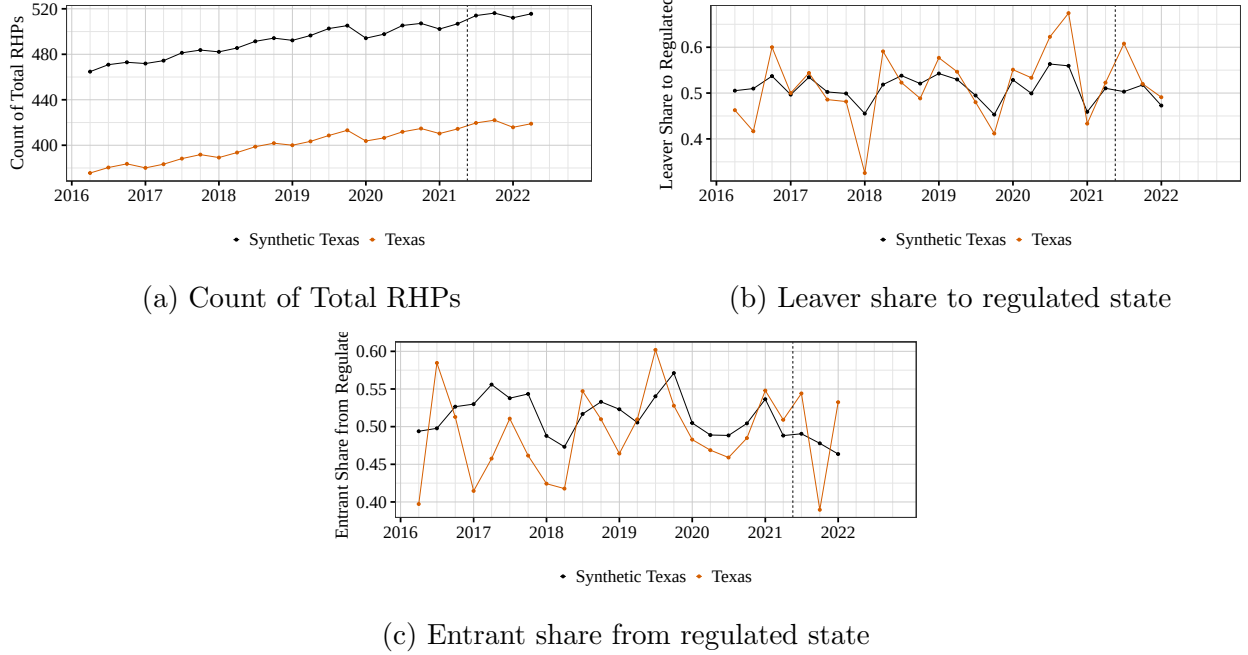


Figure 8: SDID trajectory plots for the three main-body NPI migration outcomes. Texas (orange) versus synthetic Texas (black). SDID constructs the synthetic counterfactual to match the pre-period *trend* of Texas’s RHP trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Point estimates only; see Table D1 for placebo-variance ($B = 1000$) standard errors. Control group: regulated states. Source: NPPES NPI registry.

5.2 Results: Wage

The central identification challenge in the wage analysis is that posting-level salary data are sparse and heteroskedastic: cell sizes range from a handful of postings in small states and early quarters to thousands in large states and later years. With a single treated state, this heteroskedasticity invalidates conventional cluster-robust inference. We address this using the [Ferman and Pinto \(2019\)](#) heteroskedasticity-corrected closed-form standard error, which accounts explicitly for cell-size variation. The treated group is defined consistently with the private enforcement mechanism: OB/GYN, Emergency Medicine, and OB-doing Family Medicine (Family Medicine physicians whose job title indicates obstetric, emergency, or ambiguous reproductive content).

The DDD event study for salary and posting volume (Appendix A, Figure A1) shows no post-SB-8 break in either outcome, but the pre-period coefficients are noisy: individual

quarterly estimates are not always close to zero, and a joint test rejects the hypothesis that all pre-period coefficients equal zero for both outcomes. For posted salary, the pre-period coefficients show no systematic linear trend, so the noise does not point toward a directional bias. For posting volume, the pre-period coefficients trend upward, a pattern that would bias a DDD estimator that imposes parallel pre-trends. As on the migration side, this pre-period behavior is why we do not treat the wage DDD estimates as the preferred result: the next part of this subsection applies the same SDID estimator used for migration, which does not require pre-period coefficients to be flat, and reports the corresponding SDID event study (Figure 10). The full wage DDD specification, across six fixed-effect columns and both the employment-type and geographic subsamples, is reported in Appendix A (Tables A2 and A3).

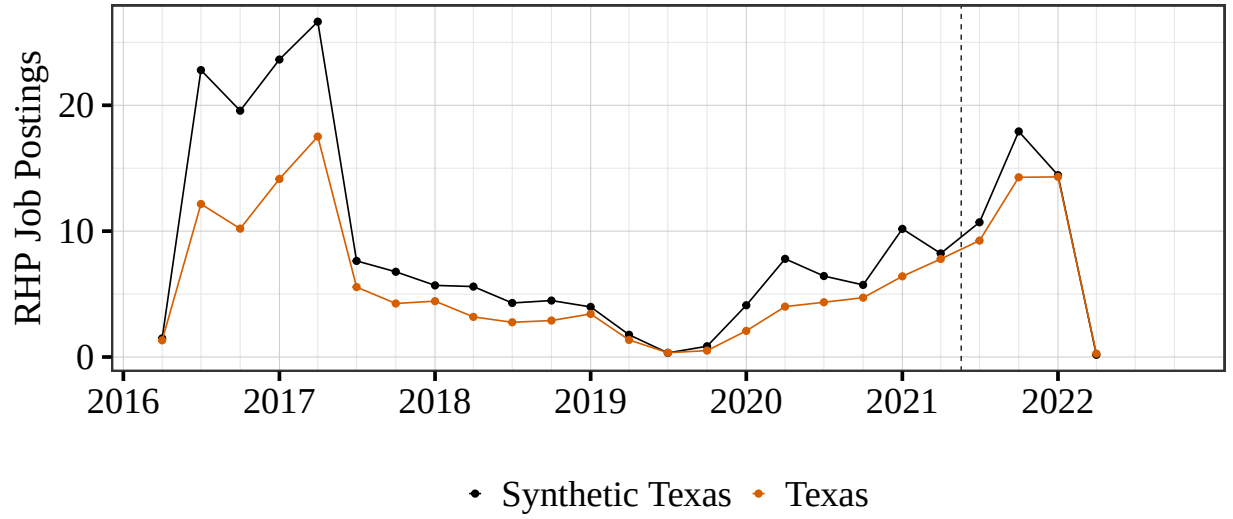
Locum Tenens Subsample. To examine a potentially more mobile segment of the physician labor market, we restrict the Wagescape sample to postings containing “locum” in the job title ($N = 25,297$ salary observations; $N = 1,182$ state-quarter cells for the posting-volume outcome, regulated-states sample). Locum tenens physicians work on temporary assignments and face lower relocation costs than permanently placed physicians, making them a potentially more responsive margin to policy shocks like SB-8. Panel C of Table A2 (Appendix A) presents triple-difference estimates for this subsample; Appendix Q reports the corresponding event studies. The salary DDD estimator is statistically indistinguishable from zero, consistent with the null in the full sample. The posting-volume DDD is positive and statistically significant, indicating that Texas reproductive health providers saw a differential increase in locum postings per 100,000 women. This pattern is directionally consistent with a supply disruption: if permanent physicians reduced their Texas presence following SB-8, employers would need to fill gaps with temporary staff. We note two threats to this interpretation. First, the posting-volume series is sensitive to thin state-quarter cells, since locum postings are a small subset of the full sample. Second, Wagescape assembles its

postings from a changing set of data partners and job boards over time, and its coverage of locum-tenens postings specifically could have expanded in Texas at a different rate than in the control states over this period. If so, a rise in Texas locum posting counts could reflect Wagescape capturing more of an existing market rather than employers posting more jobs, which would produce the same positive DDD estimate without any change in actual labor demand. We cannot distinguish this data-coverage explanation from a genuine supply response with the data available, so we interpret the posting result cautiously. Across both full-sample and locum subsamples, the wage evidence is consistent with the null migration result.

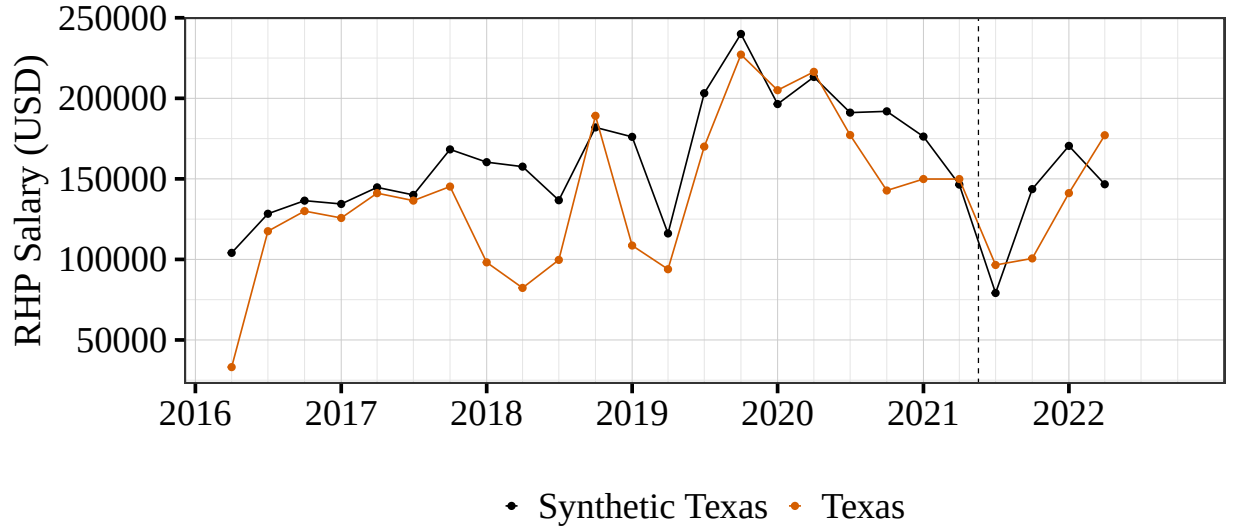
We now turn to the paper’s preferred wage estimates, mirroring Section 5.1’s treatment of migration. The noisy pre-period salary coefficients documented above are, as on the migration side, a parallel-trends violation the DDD design cannot correct for; SDID addresses it directly by reweighting the donor pool and pre-period quarters rather than assuming Texas and the control states would have moved in parallel. It operates on the RHP-only panel, with Texas as the single treated unit and the regulated states as the donor pool; inference uses the placebo variance estimator with $B = 1000$ replications. A levels-only synthetic control (SC) comparison is reported as a robustness check in Appendix E; neither outcome shows a post-signing divergence between Texas and synthetic Texas there either.

The SDID estimates are $\hat{\tau} = 0.112$ (log salary, placebo SE 0.108), $\hat{\tau} = \$9762$ (COL-adjusted salary, USD, placebo SE \$23895), and $\hat{\tau} = 1.096$ (RHP postings per 100k, placebo SE 2.450); all three are statistically indistinguishable from zero (Table D1), confirming the DDD null above under a design that does not require flat pre-period coefficients.

Figure 9 shows the SDID trajectory plots (unit weights for the log-salary outcome are shown alongside the NPI weights in Figure 7, Section 5.1). SDID is designed to match the pre-period trend of Texas, and the synthetic counterfactual tracks the actual trajectory closely in each pre-period panel. Neither outcome shows a post-signing divergence between Texas and synthetic Texas, consistent with the SC comparison (Appendix E).



(a) RHP Job Postings



(b) RHP Salary (USD)

Figure 9: SDID trajectory plots for Wagescape outcomes. Texas (orange) versus synthetic Texas (black). SDID constructs the synthetic counterfactual to match the pre-period *trend* of Texas’s wage and posting trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Some regulated states are excluded from the wage donor pool due to sparse salary coverage in early quarters. Source: Wagescape via Dewey Data Platform.

As with migration, Figure 10 extends the trajectory-plot comparison to a dynamic per-quarter SDID treatment-effect path with placebo-variance confidence intervals, following the same Ciccia (2024) construction described in Section 5.1. Pre-period coefficients are statistically indistinguishable from zero across all three wage outcomes, and no post-signing

break is evident in any panel.

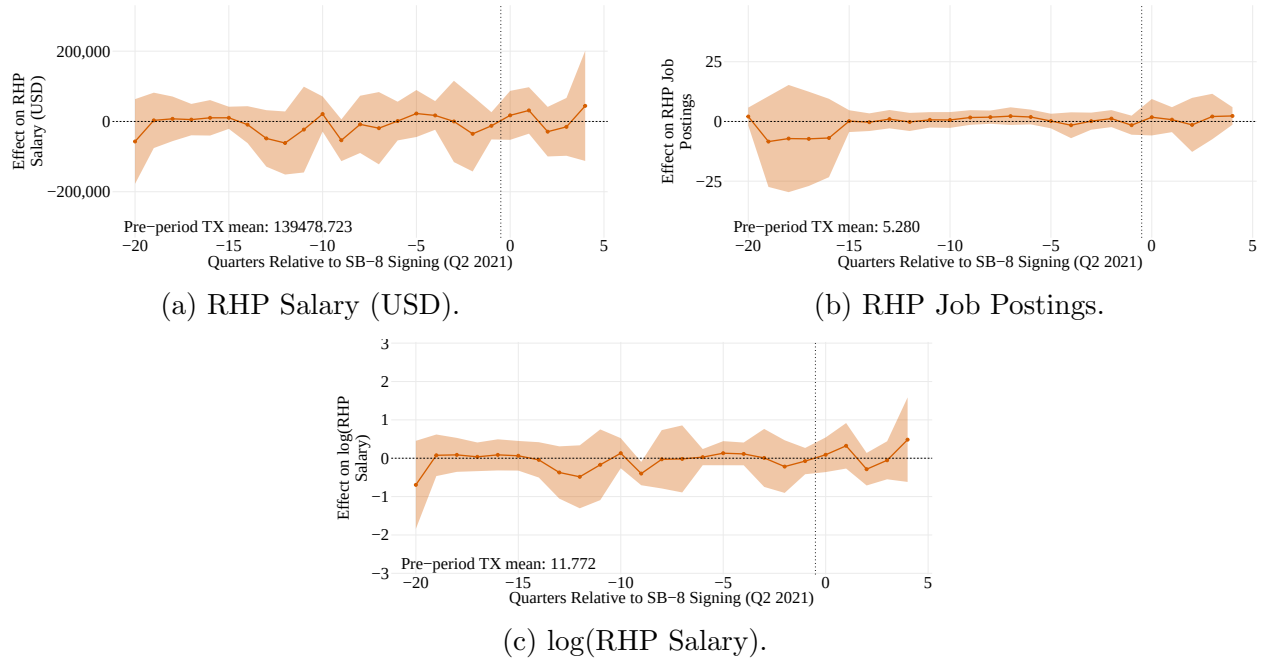


Figure 10: SDID event studies for the three Wagescape outcomes, following [Ciccia \(2024\)](#). Each point is the per-quarter SDID treatment effect (Texas versus synthetic Texas, net of the time-weighted pre-period baseline); shaded regions show 95% confidence intervals from a leave-one-out placebo procedure over the regulated-states donor pool (donor pool is smaller than for migration because some regulated states lack complete quarterly salary coverage; see Table [D1](#) notes). Quarters relative to SB-8 signing (May 19, 2021); event time 0 is Q2 2021. Control group: regulated states. Source: Wagescape via Dewey Data Platform.

Figure [11](#) collects all estimates into a single coefficient plot. Columns are active physician stock (NPI) and the three Wagescape outcomes; rows are identification strategies. For Wagescape outcomes, the upper section additionally shows the DDD estimate under five alternative fixed-effect specifications (corresponding to columns 1–5 of Tables [A2](#) and [A3](#)), from no fixed effects to the preferred unit-and-quarter-by-year specification; all DDD rows use [Ferman and Pinto \(2019\)](#) standard errors. The active stock column shows only the main DDD and synthetic estimators. Point estimates and 95% confidence intervals are centered on zero and statistically indistinguishable from zero across all outcomes and all estimators. We find no detectable change in migration, salary, or posting volume for reproductive health physicians in Texas following SB-8.^{[17](#)}

¹⁷Appendix Figure [H1](#) extends the display to all five NPI migration outcomes.

The Effect of Private Action Enforcement

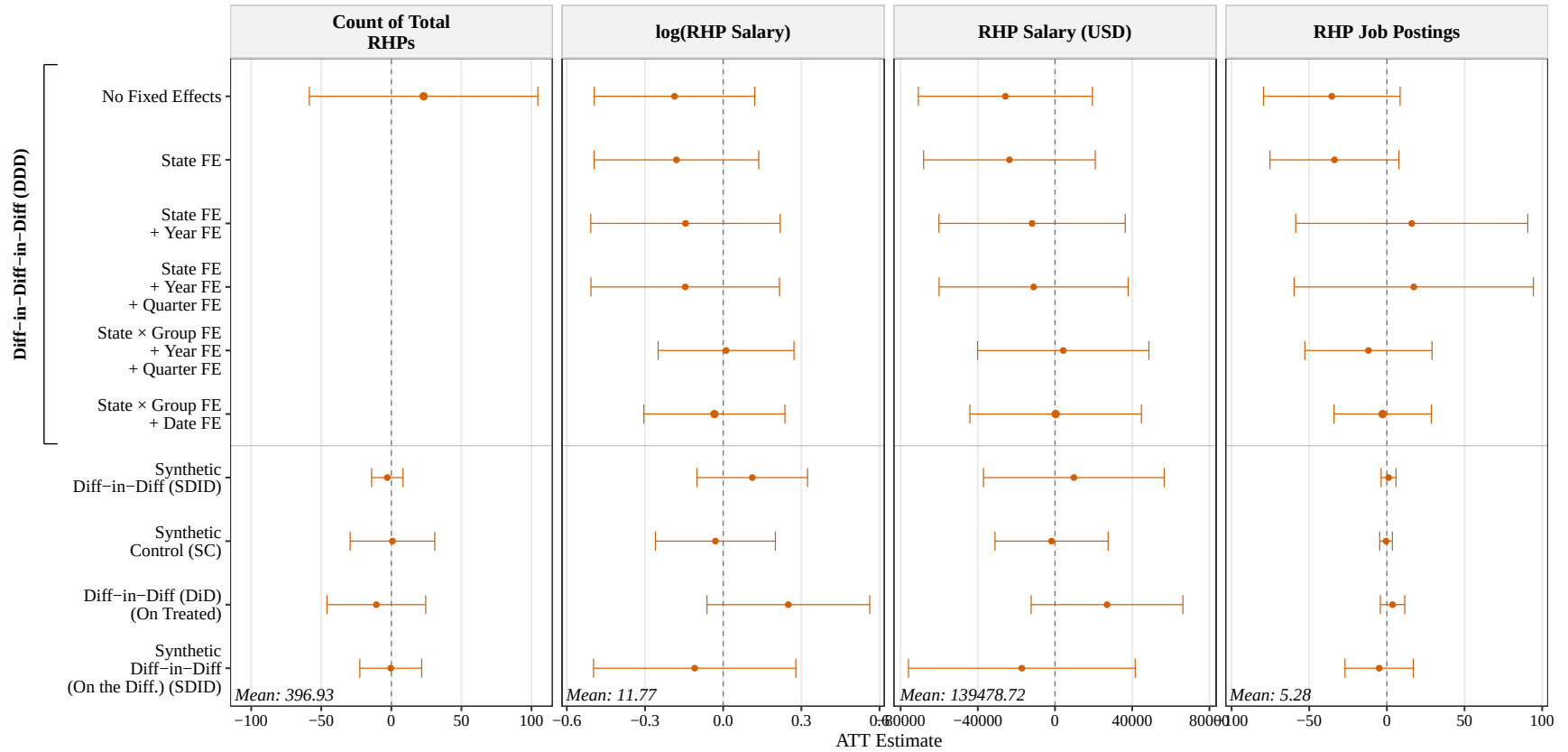


Figure 11: Coefficient plot. Columns: active physician stock (NPI, per 100k women), log posted salary, COL-adjusted salary (USD), and RHP postings per 100k (Wagescape). Rows: DDD with [Ferman and Pinto \(2019\)](#) SE (top), DDD under five alternative fixed-effect specifications for Wagescape outcomes only (all FP SE; wage FE rows are blank for the active stock column), SDID on treated physicians, SC on treated physicians, DiD on treated physicians, and SDID on the RHP – Non-RHP difference. SDID/SC/DiD use placebo variance ($B = 1000$). X-axes are centered on zero; gray italic text shows the pre-period Texas RHP mean. Control group: regulated states.

6 Discussion

This paper examines whether SB-8’s private enforcement mechanism altered the location decisions and compensation of reproductive health physicians. We rely on the synthetic difference-in-differences (SDID) estimator as our preferred identification strategy because it does not require assuming Texas and the control states would have moved in parallel absent SB-8. We find no statistically significant effect on any outcome we examine.

On the migration side, we find no differential change in exit rates or career entrances consistent with a deteriorating practice environment following SB-8’s signing on May 19, 2021. The SDID estimate for the active reproductive-health-physician stock is statistically indistinguishable from zero, as are the leaver-share, entrant-share, and new-entrant SDID estimates (Table D1); the levels-only synthetic control (SC) comparison is directionally consistent (Appendix E).

On the wage side, reproductive health providers in Texas did not command a compensating wage differential for private enforcement exposure, nor did they experience a differential wage decline. The SDID estimates are all statistically indistinguishable from zero.

The null results are consistent across eight outcomes. One interpretation is that the marginal expected cost of private enforcement exposure was insufficient to overcome the substantial financial and professional costs of interstate relocation for established physicians, estimated at \$150,000–\$250,000 in licensing, credentialing, and transition costs (see Section 3 for the sources underlying this estimate). We cannot rule out alternative explanations, including that the pool of physicians most willing to exit in response to increased operating costs had already left Texas under prior TRAP regulations, or that the threat of private enforcement litigation was too ambiguous to alter near-term location decisions.

Limitations and Next Steps

1. The retirement margin is poorly measured. Physicians do not deactivate their National Provider Identifiers when they retire, so the NPI panel substantially undercounts workforce exit. The retirement margin is particularly important for this analysis because physicians planning to retire in the near term face lower fixed costs of early exit than of relocation. As additional administrative data sources become available, including Medicare prescribing records and vital statistics restricted-access files linking birth certificates to provider identifiers, it will be possible to construct more precise measures of retirement and early career exit.
2. Heterogeneity by physician age is a natural extension. New physicians choosing an initial practice location bear no relocation cost and may be the most responsive margin; near-retirees face lower costs of early exit than of interstate relocation. As data on physician graduation year from CMS Physician Compare are linked to the NPI panel, it will be possible to examine whether younger and older cohorts exhibit differential responses.
3. County-level analysis would permit tests of whether physicians in blue Texas counties or low-relocation-cost border counties (e.g., Texarkana) are more responsive than those in rural or conservative counties, consistent with the prediction that physicians facing lower relocation costs should be more mobile.

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A Triple-Difference Estimating Equation and Full Results

The triple-difference (DDD) design referenced throughout the Results section (Table A1 and the DDD event studies) exploits three sources of variation to identify the effect of SB-8’s private enforcement mechanism: time (pre- versus post-signing, after May 19, 2021), geography (Texas versus the regulated states), and specialty (reproductive health physicians versus all other physicians). Its central identifying assumption is parallel trends: absent SB-8, Texas RHPs would have evolved at the same rate as control-state RHPs, conditional on the common trajectory of non-RHP physicians within Texas, plus no anticipation (physicians did not alter location decisions before SB-8’s signing).

$$\begin{aligned} \text{Outcome}_{ist} = & \alpha + \beta_1 \cdot \underbrace{\text{Post SB8}_t}_{=1 \text{ if after May 19, 2021}} \times \underbrace{\text{Texas}_s}_{=1 \text{ if Texas}} \times \underbrace{\text{RHP}_i}_{=1 \text{ if Reprod. Health Phys.}} \\ & + \bar{\beta} \cdot \left[\text{Post SB8}_t \times \text{Texas}_s + \text{Post SB8}_t \times \text{RHP}_i + \text{Texas}_s \times \text{RHP}_i \right. \\ & \left. + \text{Post SB8}_t + \text{Texas}_s + \text{RHP}_i \right] + \underbrace{u_{ist}}_{\text{FP SE (by state)}} \end{aligned}$$

where

$$\begin{aligned} \text{Outcome}_{ist} \in \{ & \text{Total Active, Leaver Share to Regulated, Entrant Share from Regulated,} \\ & \text{New Entrances, Retirements, Salary, and Job Posting Volume} \} \end{aligned} \quad (2)$$

The coefficient β_1 is the triple-difference estimator of interest: the differential change in outcomes for Texas RHPs, relative to non-RHP physicians in Texas and relative to RHPs in the control states, following SB-8’s signing. Non-RHP physicians absorb general Texas physician-market trends including population-driven changes in the per-100k denominator, COVID-related demand shocks, and state-level supply shifts unrelated to reproductive

health. Control-state RHPs absorb national specialty-level trends. The triple-difference structure isolates the component of any Texas RHP change that is attributable to neither confounder. The non-RHP group is defined residually (any physician not classified as OB/GYN, OB-practicing Family Medicine, or Emergency Medicine), so it includes General Practice physicians (taxonomy 208D) and Pediatricians without either group having been separately examined.¹⁸

Triple-Difference Estimates: Migration Outcomes

	Count of Total RHPs	Leaver Share to Regulated	Entrant Share from Regulated	New Entrances	Retirements
Texas $\times$ Pass-SB8	−34.876** (12.614)	0.029 (0.075)	−0.007 (0.076)	0.013 (3.114)	−0.002 (0.020)
Reproductive $\times$ Pass-SB8	−65.875*** (3.499)	−0.029 (0.021)	−0.035 (0.021)	3.196*** (0.864)	0.002 (0.006)
β_1	23.020 (41.595)	0.032 (0.045)	0.049 (0.048)	0.545 (3.230)	0.003 (0.011)
Unit FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes
Control Group	Regulated	Regulated	Regulated	Regulated	Regulated
Observations	1,404	1,349	1,348	1,404	1,404
Pre-period mean (TX RHP)	394.25	0.51	0.49	2.07	0.00

Table A1: Triple-difference estimates of the effect of SB-8 on physician outcomes per 100,000 women of childbearing age (NPI Panel, 2015–2022). Columns are the five outcome variables; the specification includes unit (state $\times$ group) and date fixed effects. β_1 is the triple-difference coefficient (Texas $\times$ RHP $\times$ Post-SB8). Ferman and Pinto (2019) heteroskedasticity-corrected SEs in parentheses. Control group: regulated states (25 states). Source: NPPES/CMS DAC.

(*) $p < 0.10$; (**) $p < 0.05$; (***) $p < 0.01$

The wage side of the same DDD design (Section 5.2) is reported across six fixed-effect specifications in Table A2 (employment-type subsamples: full sample, full-time, locum tenens) and Table A3 (geographic subsamples: full sample, urban, rural).

¹⁸Appendix M shows β_1 is essentially unchanged when both are dropped from the non-RHP pool entirely.

Wage Triple-Difference Estimates by Employment Type and Fixed-Effect Specification

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Total (Full Sample)</i>						
log(RHP Salary)	-0.187 (0.157)	-0.180 (0.161)	-0.145 (0.185)	-0.146 (0.185)	0.011 (0.133)	-0.034 (0.138)
RHP Job Postings	-35.448 (22.435)	-33.736 (21.186)	16.024 (38.114)	17.315 (39.302)	-11.864 (20.896)	-2.702 (16.002)
RHP Salary (USD)	-25730 (23004)	-23624 (22688)	-11885 (24633)	-11105 (24998)	4258 (22647)	332 (22667)
Pre-period mean (TX RHP): log(RHP Salary)			11.612			
Pre-period mean (TX RHP): RHP Job Postings			5.40			
Pre-period mean (TX RHP): RHP Salary (USD)			141042			
<i>Panel B: Full-Time (Non-Locum)</i>						
log(RHP Salary)	-0.199 (0.174)	-0.192 (0.178)	-0.156 (0.199)	-0.158 (0.197)	-0.011 (0.152)	-0.044 (0.160)
RHP Job Postings	-33.964** (15.487)	-32.660** (14.444)	7.539 (30.239)	8.585 (31.239)	-16.488 (17.566)	-9.070 (11.812)
RHP Salary (USD)	-28981 (26196)	-26861 (25955)	-15286 (27697)	-14985 (27780)	-1286 (25894)	-2384 (26062)
Pre-period mean (TX RHP): log(RHP Salary)			11.599			
Pre-period mean (TX RHP): RHP Job Postings			4.84			
Pre-period mean (TX RHP): RHP Salary (USD)			139654			
<i>Panel C: Locum Tenens</i>						
log(RHP Salary)	0.132 (0.154)	0.130 (0.156)	0.118 (0.150)	0.125 (0.160)	0.178 (0.206)	0.045 (0.168)
RHP Job Postings	-0.503 (9.978)	-0.255 (9.819)	8.950 (8.559)	9.431 (8.704)	5.062 (8.611)	7.120 (8.008)
RHP Salary (USD)	39633 (43170)	40104 (42729)	44243 (29112)	49438* (25228)	60117* (33377)	37285 (27623)
Pre-period mean (TX RHP): log(RHP Salary)			12.165			
Pre-period mean (TX RHP): RHP Job Postings			0.72			
Pre-period mean (TX RHP): RHP Salary (USD)			199383			
State FE	No	Yes	Yes	Yes	—	—
Year FE	No	No	Yes	Yes	Yes	—
Quarter-of-Year FE	No	No	No	Yes	Yes	—
State $\times$ Group FE	No	No	No	No	Yes	Yes
Quarter $\times$ Year FE	No	No	No	No	No	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes	Yes

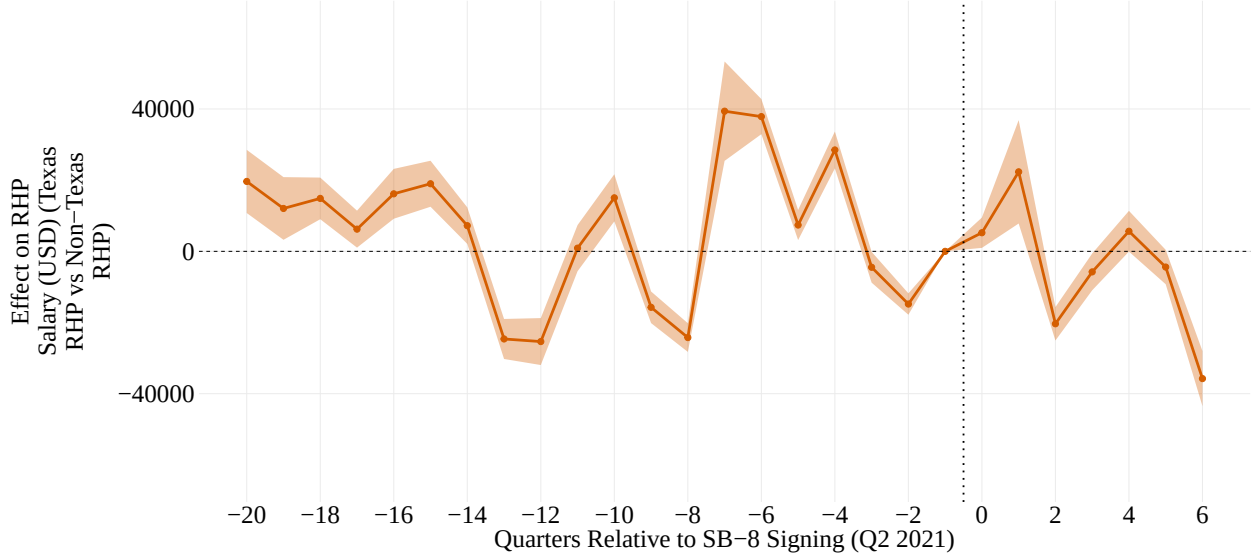
Table A2: Wage triple-difference β_1 estimates across employment-type subsamples and six fixed-effect specifications (columns). Panel A is the full sample; Panel B restricts to full-time (non-locum) postings; Panel C shows locum tenens postings only (identified by “locum” in job title, $\approx 10\%$ of postings). RHP Salary (USD) is BEA RPP cost-of-living-adjusted posted salary; RHP Job Postings is postings per 100,000 women of childbearing age. Columns (5)–(6) use state $\times$ group (RHP/non-RHP) unit fixed effects. All regressions are unweighted OLS. [Ferman and Pinto \(2019\)](#) heteroskedasticity-corrected SEs (M = posting count). Control group: regulated states (25 states).

Wage Triple-Difference Estimates by Geography and Fixed-Effect Specification

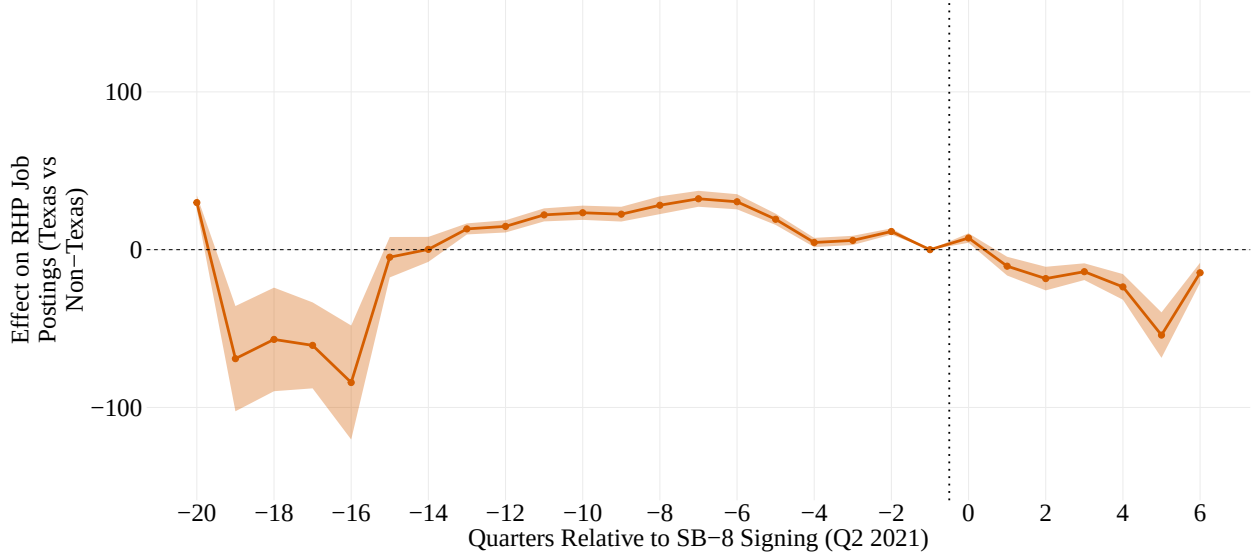
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Total (Full Sample)</i>						
log(RHP Salary)	-0.187 (0.157)	-0.180 (0.161)	-0.145 (0.185)	-0.146 (0.185)	0.011 (0.133)	-0.034 (0.138)
RHP Job Postings	-35.448 (22.435)	-33.736 (21.186)	16.024 (38.114)	17.315 (39.302)	-11.864 (20.896)	-2.702 (16.002)
RHP Salary (USD)	-25730 (23004)	-23624 (22688)	-11885 (24633)	-11105 (24998)	4258 (22647)	332 (22667)
Pre-period mean (TX RHP): log(RHP Salary)				11.612		
Pre-period mean (TX RHP): RHP Job Postings				5.40		
Pre-period mean (TX RHP): RHP Salary (USD)				141042		
<i>Panel B: Urban</i>						
log(RHP Salary)	-0.144 (0.289)	-0.137 (0.294)	-0.093 (0.328)	-0.093 (0.327)	-0.013 (0.198)	-0.038 (0.192)
RHP Job Postings	-3144*** (705)	-3155*** (712)	-2710*** (534)	-2697*** (533)	-2035*** (598)	-1964*** (569)
RHP Salary (USD)	-16849 (31167)	-15072 (31999)	-3564 (38868)	-3203 (39029)	1836 (28605)	1673 (28416)
Pre-period mean (TX RHP): log(RHP Salary)				11.541		
Pre-period mean (TX RHP): RHP Job Postings				213.19		
Pre-period mean (TX RHP): RHP Salary (USD)				132593		
<i>Panel C: Rural</i>						
log(RHP Salary)	-0.314* (0.170)	-0.299* (0.171)	-0.307* (0.175)	-0.324* (0.174)	-0.275 (0.184)	-0.341** (0.170)
RHP Job Postings	-139 (105)	-140 (106)	-73.430 (55.461)	-71.764 (55.288)	-43.244 (83.021)	-37.312 (75.410)
RHP Salary (USD)	-58974 (40785)	-55940 (40158)	-51546 (39626)	-53509 (39621)	-46241 (40608)	-51602 (39832)
Pre-period mean (TX RHP): log(RHP Salary)				11.921		
Pre-period mean (TX RHP): RHP Job Postings				22.90		
Pre-period mean (TX RHP): RHP Salary (USD)				177442		
State FE	No	Yes	Yes	Yes	—	—
Year FE	No	No	Yes	Yes	Yes	—
Quarter-of-Year FE	No	No	No	Yes	Yes	—
State $\times$ Group FE	No	No	No	No	Yes	Yes
Quarter $\times$ Year FE	No	No	No	No	No	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes	Yes

Table A3: Wage triple-difference β_1 estimates across geographic subsamples and six fixed-effect specifications (columns). Panel A is the full sample; Panels B and C split by posting zip code urbanicity (USDA RUCA 2020); zip codes with no RUCA match ($\approx 30\%$ of postings) are excluded from Panels B–C. RHP Salary (USD) is BEA RPP cost-of-living-adjusted posted salary. RHP Job Postings in Panel A is postings per 100,000 women of childbearing age; in Panels B–C it is raw posting count (levels), since the per-100k denominator is state-level and does not vary by urbanicity. Columns (5)–(6) use state $\times$ group (RHP/non-RHP) unit fixed effects. [Ferman and Pinto \(2019\)](#) heteroskedasticity-corrected SEs (M = posting count). Control group: regulated states (25 states).

The DDD event study underlying the discussion in Section 5.2 is shown below (Figure A1); the paper’s preferred SDID event study for the same outcomes is Figure 10 in the main text.



(a) Physician salary in USD: Texas RHP vs. non-Texas RHP triple-difference, quarterly.



(b) RHP job postings per 100k women: Texas vs. non-Texas triple-difference, quarterly.

Figure A1: Triple-difference event study estimates for wage and posting outcomes using Wagescape data. Coefficients β_k are the quarterly treatment effects ($\text{TX} \times \text{RHP} \times \text{Post}$ interaction) relative to Q1 2021 (reference period: event time -1). Panel (a) uses BEA RPP cost-of-living adjusted posted salary in USD as the outcome; panel (b) uses treated RHP job postings per 100,000 women of childbearing age. Unit (state $\times$ group) and quarter-year fixed effects are included. Sample: Wagescape physician job postings, 2016–2022, regulated states. Shaded regions show 95% confidence intervals. Pre-period coefficients are noisy, with a significant upward pre-trend in posting volume; see Section 5.2 for discussion. The post-SB-8 period shows no break in either salary or posting volume. Cluster-robust SEs clustered by state. Source: Wagescape via Dewey Data Platform.

B OB/GYN-Only Triple-Difference: Migration Results

The main-text RHP treatment group pools OB/GYN, Family Medicine physicians with an OB-practicing title, and Emergency Medicine. As a robustness check on this specialty definition, we re-estimate the same triple-difference specification restricting the treated group to OB/GYN alone (SOC-equivalent taxonomy 207V), against the same non-OB/GYN comparison group and regulated-states control group used throughout. Table B1 reports the scalar DDD estimates; Figure B2 shows the corresponding DDD event study, and Figure B1 shows the underlying Texas-vs.-regulated-states DiD for OB/GYN physicians alone. As in the main RHP results, no outcome shows a statistically significant post-signing break.

Triple-Difference Estimates: OB/GYN-Only Migration Outcomes

	Count of Total OB/GYN	Leaver Share to Regulated	Entrant Share from Regulated	New Entrances	Retirements
Texas $\times$ Pass-SB8	-44.949* (19.998)	0.041 (0.111)	0.000 (0.119)	0.598 (5.450)	0.001 (0.023)
OB/GYN $\times$ Pass-SB8	-129.536*** (5.546)	-0.011 (0.032)	0.024 (0.035)	7.851*** (1.512)	0.010 (0.006)
β_1	43.166 (52.751)	-0.038 (0.144)	0.052 (0.078)	-0.625 (5.575)	-0.002 (0.022)
Unit FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Ferman-Pinto SE	Yes	Yes	Yes	Yes	Yes
Control Group	Regulated	Regulated	Regulated	Regulated	Regulated
Observations	1,404	1,247	1,253	1,404	1,404
Pre-period mean (TX OB/GYN)	85.17	0.52	0.44	0.32	0.00

Table B1: Triple-difference estimates restricting the treated group to OB/GYN physicians only (excludes Family Medicine and Emergency Medicine), per 100,000 women of childbearing age (NPI Panel, 2015–2022). Specification and standard errors as in Table A1. Control group: regulated states (25 states). Source: NPPES/CMS DAC.

(*) $p < 0.10$; (**) $p < 0.05$; (***) $p < 0.01$

OB/GYN Physicians in Texas vs. Regulated States (DiD)

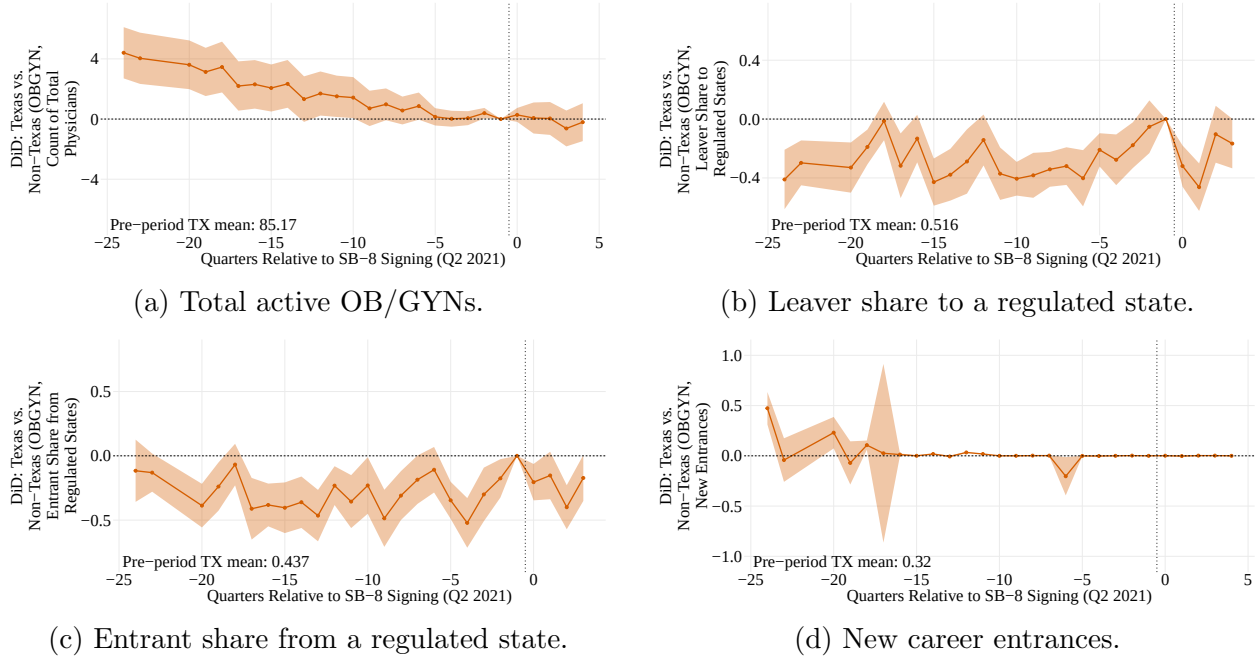


Figure B1: Difference-in-differences event study estimates, Texas versus regulated-states OB/GYN physicians (no Family Medicine or Emergency Medicine). Coefficients β_k are quarterly treatment effects (Texas $\times$ event time) relative to Q1 2021 (reference period: event time -1). Leaver/entrant share are shares (0–1) of interstate movers. State and date fixed effects are included. Sample: 2015–2022 quarterly, regulated states. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

OB/GYN vs. Non-OB/GYN Physicians in Texas vs. Regulated States (DDD)

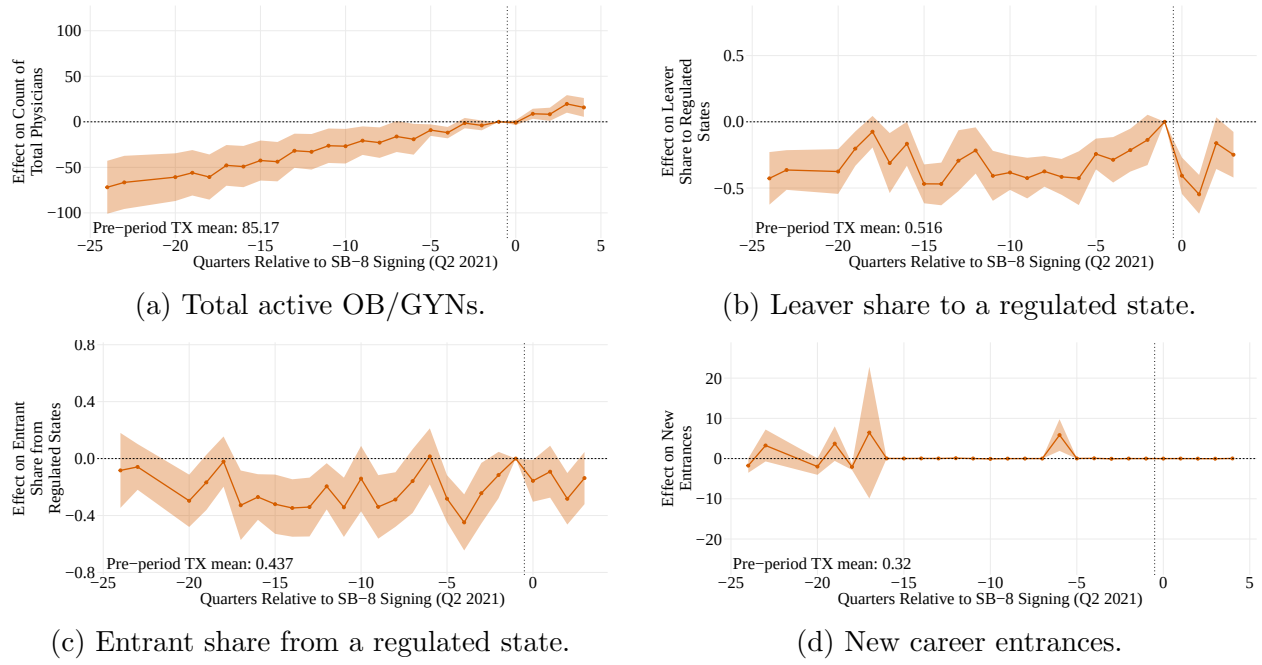


Figure B2: Triple-difference event study estimates restricting the treated group to OB/GYN physicians only. The estimating equation mirrors Figure 5, replacing the RHP indicator with an OB/GYN indicator. Leaver/entrant share are shares (0–1) of interstate movers. Unit (state $\times$ group) and date fixed effects are included. Sample: 2015–2022 quarterly, regulated states. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: NPPES/CMS DAC.

C Dobbs-Extension Robustness Check

The main analysis stops at the last full quarter before the June 24, 2022 Supreme Court decision in *Dobbs v. Jackson Women’s Health Organization*, which superseded SB-8’s abortion restrictions with a broader ban. As a robustness check on whether the null result holds just past that cutoff, this appendix re-estimates the same main-spec estimators (DiD-FP, SDID, SC; regulated-states sample) with one additional quarter (Q3 2022, July–September) added back to the sample. This is a check on the headline estimates only, not a re-run of every robustness exercise in the paper.

Figure C1 plots the active RHP stock and the three wage outcomes; Table C1 reports all eight outcomes, including the four migration outcomes (leaver share, entrant share, new career entrances, and retirements) not shown in the figure. Every outcome in both remains statistically indistinguishable from zero on the extended window, consistent with the main-text conclusion.

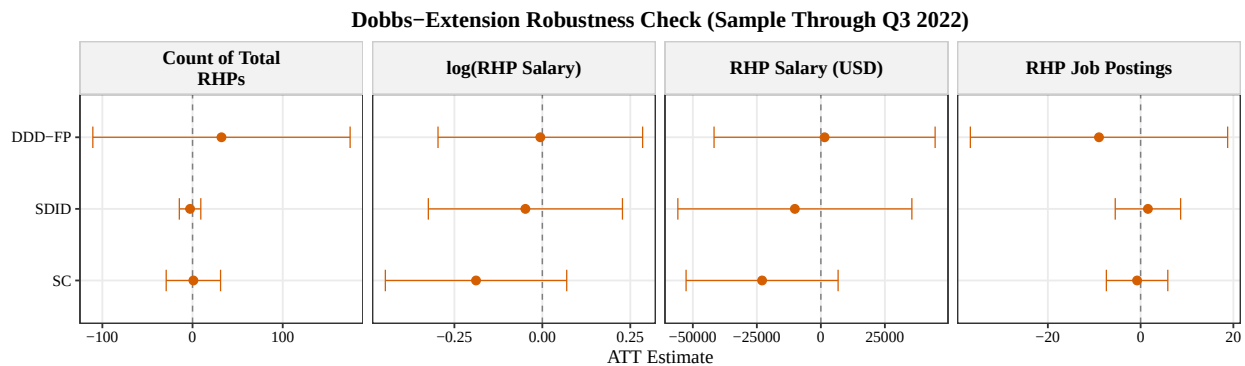


Figure C1: Main-spec estimates (DiD-FP, SDID, SC) for the active RHP stock and the three wage outcomes, re-estimated on the sample extended one quarter past the pre-Dobbs cutoff (through 2022-07-01, i.e. including Q3 2022). Regulated-states sample. Error bars show 95% confidence intervals (placebo variance for SDID/SC, [Ferman and Pinto \(2019\)](#) for DiD-FP). Source: NPPES/CMS DAC and Wagescape.

Dobbs-Extension Robustness Check: Sample Through Q3 2022

Outcome	SDID	SC	DDD-FP
<i>Panel A: NPI Migration Outcomes</i>			
Count of Total RHPs	-2.747 (6.074)	0.929 (15.379)	32.088 (72.778)
Leaver Share to Regulated	0.040 (0.075)	0.001 (0.046)	0.031 (0.041)
Entrant Share from Regulated	0.056 (0.106)	0.041 (0.073)	0.055 (0.043)
New Entrances	0.005 (0.019)	0.012 (0.015)	-0.055 (5.933)
Retirements	-0.003 (0.000)	-0.000 (0.001)	-0.002 (0.069)
<i>Panel B: Wagescape Wage Outcomes</i>			
log(RHP Salary)	-0.048 (0.141)	-0.188 (0.132)	-0.006 (0.148)
RHP Job Postings	1.578 (3.596)	-0.761 (3.379)	-8.973 (14.151)
RHP Salary (USD)	-10,155 (23,313)	-22,951 (15,145)	1,456 (22,024)
Robustness check extending the main sample one additional quarter past the pre-Dobbs cutoff (through Q3 2022, i.e. 2022-07-01). Placebo variance for SDID and SC ($B = 1,000$). DDD-FP: main-spec (unit + date FE) Ferman–Pinto (2019) closed-form SE, same specification as Table A1 (all five NPI outcomes, including Leaver/Entrant Share) / Table A2 but re-estimated on the extended window. SEs in parentheses.			

Table C1: **Dobbs-extension robustness check.** Same estimators and specification as Table D1, re-estimated with the sample extended one additional quarter past the pre-Dobbs cutoff (through 2022-07-01). Regulated-states sample.

D Synthetic Estimator Results

Table [D1](#) reports point estimates and standard errors for the SDID, SC, and DiD estimators alongside the preferred DDD estimates. Inference for the synthetic estimators uses the placebo variance estimator with $B = 1000$ replications.

SDID, SC, and DiD Estimates

Outcome	SDID	SC	DDD
<i>Panel A: NPI Migration Outcomes</i>			
Count of Total RHPs	-2.877	0.793	23.020
	(5.729)	(15.400)	(41.595)
Leaver Share to Regulated	0.049	0.011	0.032
	(0.092)	(0.071)	(0.045)
Entrant Share from Regulated	0.045	0.054	0.049
	(0.144)	(0.103)	(0.048)
New Entrances	0.000	0.010	0.545
	(0.012)	(0.018)	(3.230)
Retirements	-0.003	-0.001	0.003
	(0.000)	(0.001)	(0.011)
<i>Panel B: Wagescape Wage Outcomes</i>			
log(RHP Salary)	0.112	-0.030	-0.034
	(0.108)	(0.118)	(0.138)
RHP Job Postings	1.096	-0.540	-2.702
	(2.450)	(2.088)	(16.002)
RHP Salary (USD)	9,762	-1,792	332
	(23,895)	(14,948)	(22,667)

Table D1: SDID, SC, and DiD estimates for all eight outcomes (five NPI migration + three Wagescape wage) alongside the preferred DDD estimates from Tables A1 and A2. Count of Total RHPs, New Entrances, and Retirements are per 100,000 women of childbearing age; Leaver/Entrant Share are shares (0–1) of interstate movers landing in or arriving from a regulated state. SDID, SC, and DiD operate on the RHP-only panel with Texas as the single treated unit; inference via placebo variance ($B = 1000$). DDD uses Ferman and Pinto (2019) heteroskedasticity-corrected SEs on the full RHP + non-RHP panel. Control group: regulated states. SEs in parentheses.

E Synthetic Control Trajectory Plots

The Synthetic Control (SC) estimator weights donor states to match Texas’s pre-period outcome *levels*, without time-weighting; this is a simpler counterfactual than SDID, which additionally reweights pre-period quarters to match Texas’s pre-period *trend* (Section 5.1). We report SC alongside SDID throughout the paper (Table D1); this appendix shows the underlying trajectory plots for both the migration and Wagescape outcomes.

Figure E1 shows the SC trajectory plots for NPI outcomes: Texas (solid orange) against synthetic Texas (dashed black). No NPI outcome shows a post-signing divergence between Texas and synthetic Texas.

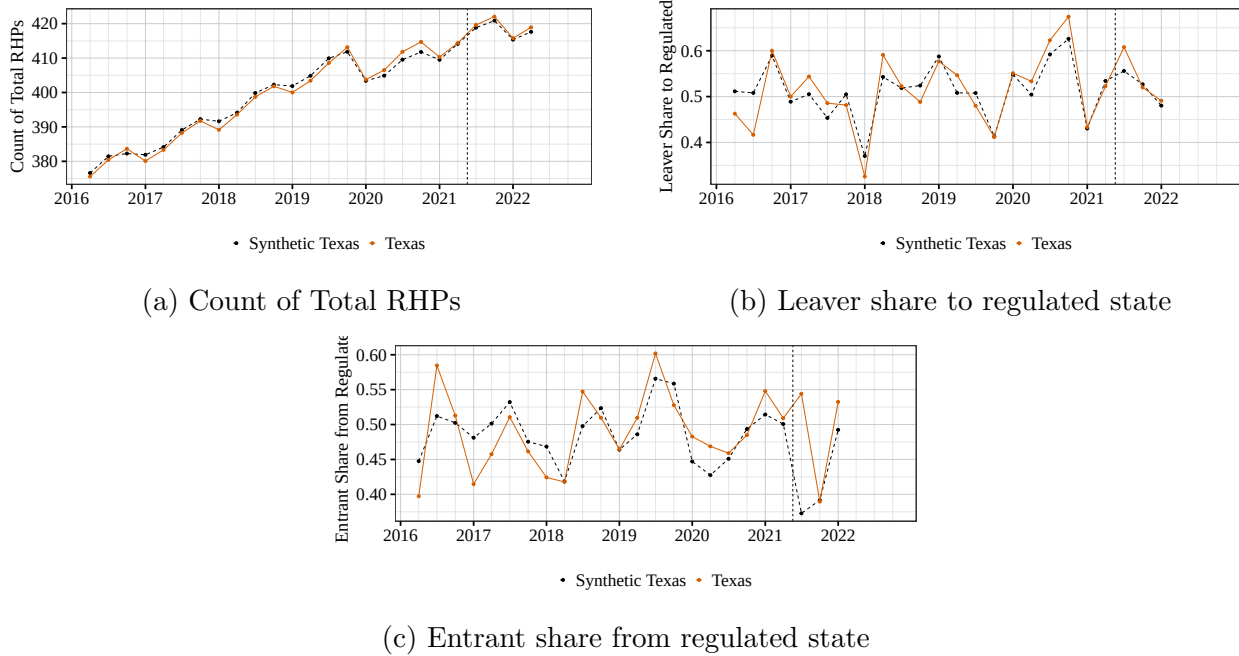
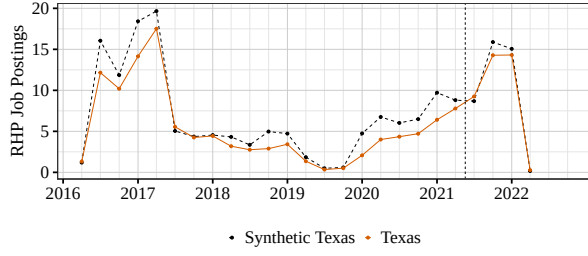
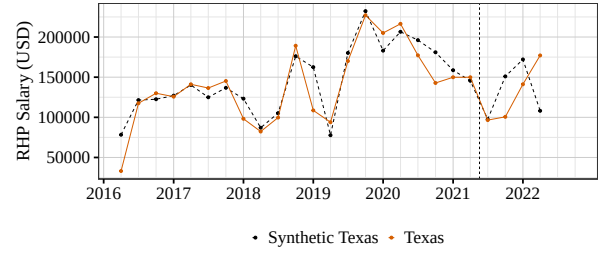


Figure E1: Synthetic control (SC) trajectory plots for NPI outcomes. Texas (solid orange) versus synthetic Texas (dashed black). SC constructs the synthetic counterfactual to match the pre-period *level* of Texas’s RHP trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Control group: regulated states. Source: NPPES NPI registry.

Figure E2 shows the SC trajectory plots for the Wagescape outcomes. Neither outcome shows a post-signing divergence between Texas and synthetic Texas.



(a) RHP postings per 100k



(b) COL-adjusted salary (USD)

Figure E2: Synthetic control (SC) trajectory plots for Wagescape outcomes. Texas (solid orange) versus synthetic Texas (dashed black). SC constructs the synthetic counterfactual to match the pre-period *level* of Texas's wage and posting trajectory. Dashed vertical line marks SB-8 signing (May 19, 2021). Control group: regulated states. Source: Wagescape via Dewey Data Platform.

F SDID on the RHP – NonRHP Difference

As an additional robustness check, we apply SDID to the within-state, within-quarter difference between RHP and non-RHP outcomes. For each state s and quarter t , we compute $\Delta_{st} = y_{st}^{\text{RHP}} - y_{st}^{\text{NonRHP}}$ and then apply SDID with Texas as the single treated unit and the Dench-Lindo control states as the donor pool. This estimator replicates the triple-difference logic in the SDID framework: rather than netting out the non-RHP trend via an explicit interaction term, it does so by pre-differencing the outcome before constructing the synthetic counterfactual. Under the identifying assumption that the within-state RHP-minus-NonRHP gap would have evolved in parallel across states absent SB-8, this estimator recovers the same causal parameter as the DDD β_1 . Inference uses placebo variance with $B = 1000$ replications. Figures F1 and F3 show trajectory plots; all estimates are null, consistent with the main results.

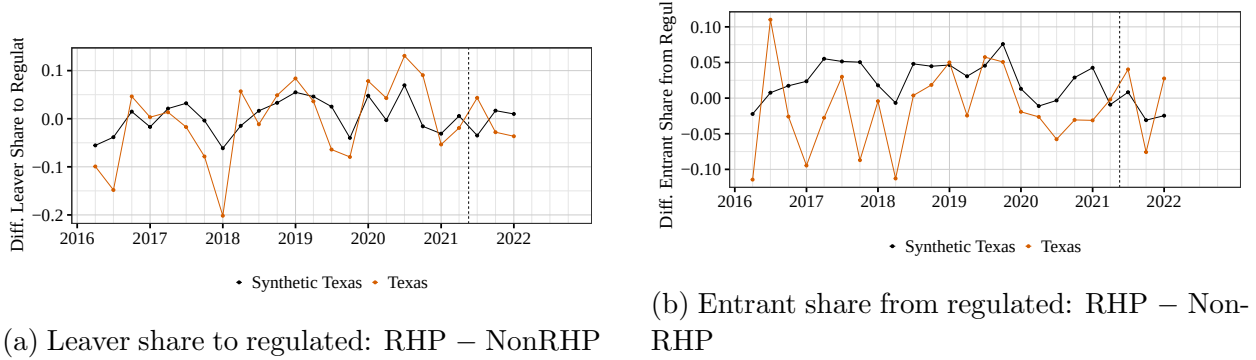
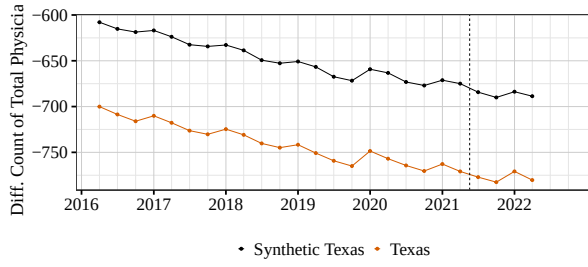
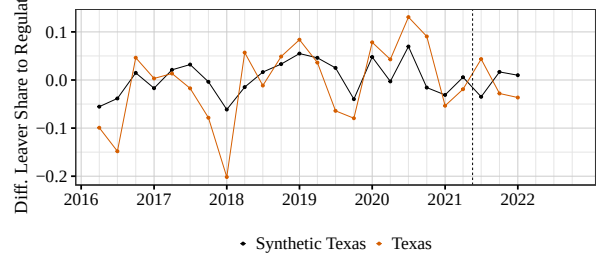


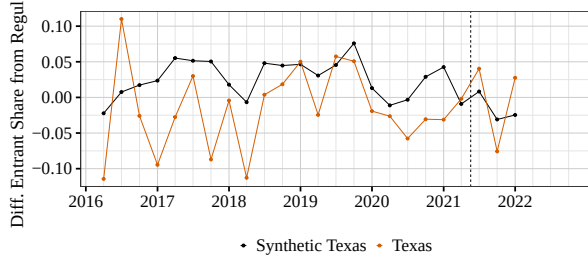
Figure F2: SDID trajectory plots on the (RHP – NonRHP) difference for the intensive migration margin. Texas (solid orange) versus synthetic Texas (dashed black). The outcome is the within-state RHP-minus-NonRHP difference in the leaver/entrant destination share, pre-differenced before applying SDID. Dashed vertical line marks SB-8 signing (May 19, 2021). Point estimates only. Control group: regulated states. Source: NPPES NPI registry.



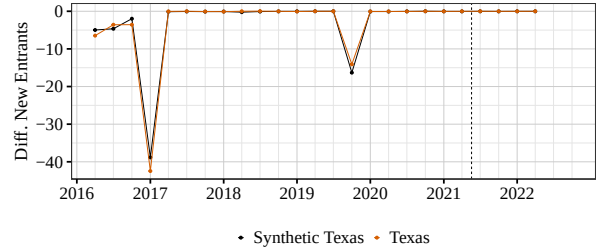
(a) Active: RHP – NonRHP per 100k



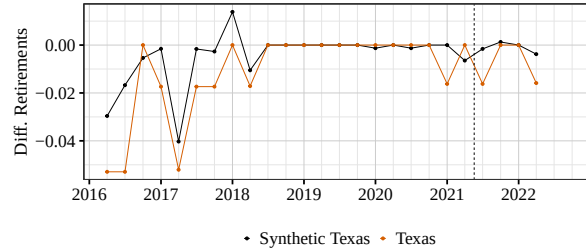
(b) Leaver share to regulated state: RHP – NonRHP



(c) Entrant share from regulated state: RHP – NonRHP

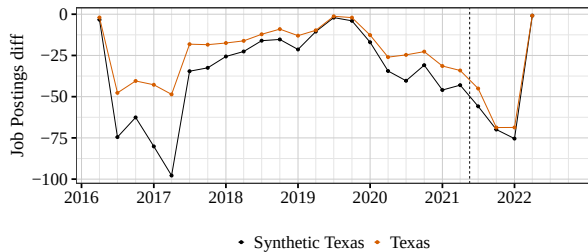


(d) New entrants: RHP – NonRHP per 100k

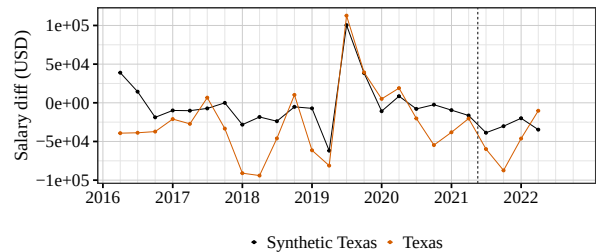


(e) Retirements: RHP – NonRHP per 100k

Figure F1: SDID trajectory plots on the (RHP – NonRHP) difference for NPI outcomes. Texas (solid orange) versus synthetic Texas (dashed black). The outcome is the within-state difference between RHP and non-RHP physician counts per 100,000 women of childbearing age, pre-differenced before applying SDID. Dashed vertical line marks SB-8 signing (May 19, 2021). Point estimates only. Control group: regulated states. Source: NPPES NPI registry.



(a) RHP Job Postings: RHP – NonRHP



(b) RHP Salary (USD): RHP – NonRHP

Figure F3: SDID trajectory plots on the (RHP – NonRHP) difference for Wagescape outcomes. Texas (solid orange) versus synthetic Texas (dashed black). The outcome is the within-state RHP-minus-NonRHP difference in posting volume or COL-adjusted salary, pre-differenced before applying SDID. Dashed vertical line marks SB-8 signing (May 19, 2021). Point estimates only. Control group: regulated states. Source: Wagescape via Dewey Data Platform.

G Cross-State OB/GYN Posting Share Growth

Section 4.4 notes that Texas’s OB/GYN share of total Wagescape physician postings rose over the sample period. Figure G1 ranks all 51 states (50 states plus DC) by the growth in this share, comparing the pre-2021 mean to the post-2021 mean. Texas ranks 16th of 51, with several states (e.g., Tennessee, Utah, Missouri) showing comparable or larger growth. This is consistent with the growth reflecting broad-based Wagescape coverage expansion documented in Panel (b) of Figure 2, rather than a Texas-specific, policy-driven compositional shift in job postings.

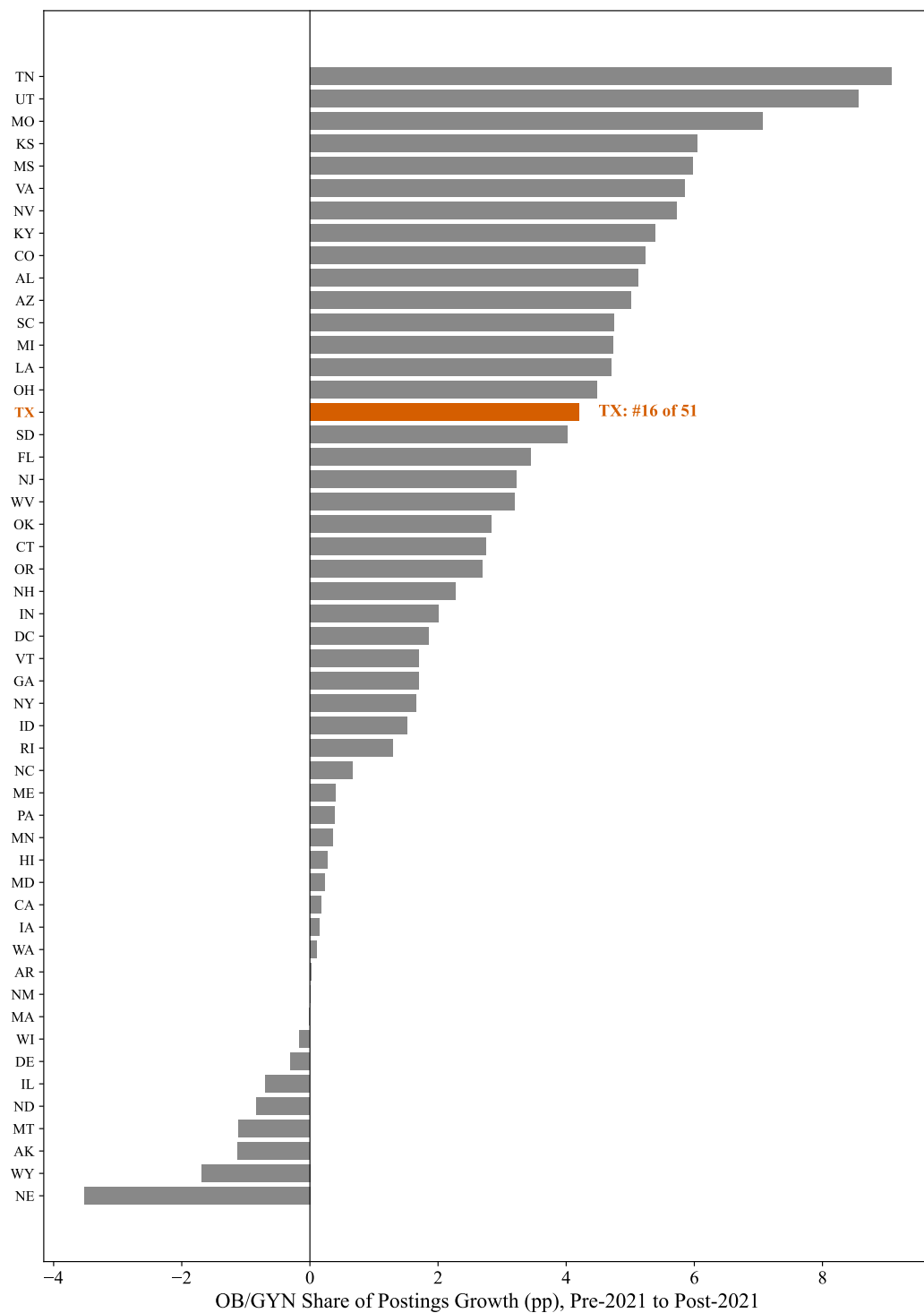


Figure G1: OB/GYN share of total physician postings, growth from pre-2021 mean to post-2021 mean, by state. Texas (orange) ranks 16th of 51. Source: Wagescape via Dewey Data Platform.

H Full NPI Migration Coefficient Plot

Figure [H1](#) replicates the structure of Figure [11](#) for all five NPI migration outcomes. Each column is one outcome; rows are the same set of identification strategies. All estimates are null.

NPI Migration Outcomes: Estimates Across Methods (Regulated States)

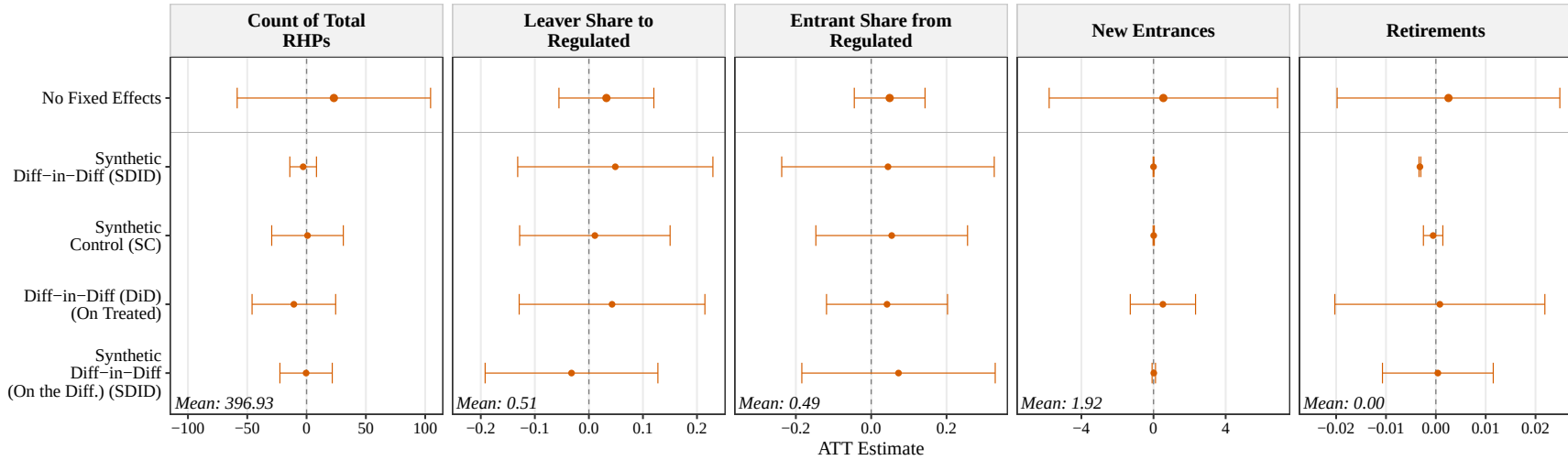


Figure H1: Coefficient plot for all five NPI migration outcomes. Columns: active stock (per 100,000 women of childbearing age), leaver share to a regulated state, entrant share from a regulated state (both shares, 0–1, of interstate movers), new career entrants, and retirements (both per 100,000 women of childbearing age). The rightmost two columns should be interpreted with caution: measurement error currently prevents precise estimation or interpretation of these coefficients. Rows: DDD (Ferman and Pinto (2019) SE), SDID on treated physicians, SC on treated physicians, DiD on treated physicians, and SDID on the (RHP – Non-RHP) difference. SDID/SC/DiD use placebo variance ($B = 1000$). X-axes are centered on zero; gray italic text shows the pre-period Texas RHP mean. Control group: regulated states.

I HonestDiD Sensitivity Analysis

The event-study designs throughout this paper assume exact parallel trends: absent SB-8, Texas would have continued on the same trajectory as the control states. [Rambachan and Roth \(2023\)](#)’s HonestDiD approach relaxes this assumption and asks how large a post-treatment deviation from parallel trends (relative to the deviation already visible in the pre-period coefficients) the paper’s conclusions could tolerate before flipping. We apply it here as a check on the event-study specifications specifically (the same saturated triple-interaction formula used throughout Section 5 and Section 5.2), not as a replacement for the collapsed-panel Ferman–Pinto inference used in the paper’s main tables: with a single treated unit, HonestDiD’s cluster-robust asymptotics and the main tables’ closed-form small-sample correction are built for different settings, so the two should not be read as restating the same number under a different method.

We use the “relative magnitudes” restriction of [Rambachan and Roth \(2023\)](#): the post-treatment bias is bounded by $\bar{M}$ times the largest observed pre-period coefficient (in absolute value). $\bar{M} = 0$ recovers the exact-parallel-trends case; larger $\bar{M}$ allows proportionally larger post-treatment violations. The target parameter is the uniform average of the post-treatment event-study coefficients, following the construction in [Rambachan and Roth \(2023\)](#)’s own applications. Table II reports, for each outcome, the point estimate, the exact-parallel-trends (“Original,” $\bar{M} = 0$) confidence interval, the robust confidence interval at $\bar{M} = 1$ and $\bar{M} = 2$, and the breakdown value $\bar{M}^*$, the smallest $\bar{M}$ at which the robust interval first includes zero, reported only when the Original interval itself excludes zero.

Three of the five migration outcomes (Active Physicians, Leaver Share, Entrant Share) already have an Original confidence interval that includes zero, consistent with the main-text null result. New Entrances, Retirements, and both Wagescape outcomes have an Original interval that excludes zero, but each has a small breakdown value ($\bar{M}^* = 0.25$): a post-treatment violation of parallel trends only one-quarter the size of the largest pre-period deviation already observed is enough to bring zero back into the confidence interval. We read

this as reinforcing the paper’s null conclusion rather than complicating it: the handful of outcomes with a nominally significant Original interval are not robust to a modest, plausible amount of the same kind of pre-trend violation the event studies already exhibit.¹⁹

HonestDiD Sensitivity Analysis: Migration and Wagescape Outcomes

Outcome	Point Est.	Original CI	Robust CI ($\bar{M} = 1$)	Robust CI ($\bar{M} = 2$)	Breakdown $\bar{M}^*$
<i>Panel A: Migration (NPI, regulated states)</i>					
Active Physicians	3.73	[-1.08, 8.55]	[-30.95, 38.13]	[-49.15, 49.15]	–
New Entrances	0.02	[0.00, 0.04]	[-0.18, 0.18]	[-0.18, 0.18]	0.25
Retirements	0.01	[0.01, 0.01]	[-0.02, 0.02]	[-0.02, 0.02]	0.25
Leaver Share to Regulated	0.09	[-0.04, 0.23]	[-0.91, 1.07]	[-1.35, 1.35]	–
Entrant Share from Regulated	0.05	[-0.02, 0.13]	[-0.69, 0.77]	[-0.77, 0.77]	–
<i>Panel B: Wagescape (regulated states)</i>					
RHP Salary (COL-Adjusted, USD)	-4723.78	[-7822.61, -1624.95]	[-31621.29, 31621.29]	[-31621.29, 31621.29]	0.25
RHP Job Postings (per 100k women)	-18.24	[-23.98, -12.51]	[-58.52, 58.52]	[-58.52, 58.52]	0.25

Table I1: HonestDiD sensitivity bounds (Rambachan and Roth, 2023, relative magnitudes restriction) for the event-study specifications used throughout the paper, regulated-states sample. Point estimate and confidence intervals are the uniform average of the post-treatment event-study coefficients. “Original” is the exact-parallel-trends ($\bar{M} = 0$) confidence interval; robust CIs allow a post-treatment violation up to $\bar{M}$ times the largest observed pre-period coefficient. Breakdown $\bar{M}^*$ is the smallest $\bar{M}$ at which the robust CI includes zero, reported only when the Original CI excludes zero. Source: NPPES/CMS DAC; Wagescape via Dewey Data Platform.

Figure I1 shows the full sensitivity curve for the active RHP stock, the paper’s primary migration outcome, as an example of how the robust confidence interval widens with $\bar{M}$.

¹⁹Several outcomes’ robust intervals reach an identical width at $\bar{M} = 0.5$ and stay there through $\bar{M} = 2$ (visible in Figure I1 and Table I1’s $\bar{M} = 1$ versus $\bar{M} = 2$ columns). We have not fully diagnosed whether this reflects a genuine feature of the relative-magnitudes bound for these outcomes or a numerical grid-boundary artifact of the default sensitivity grid; it does not affect the breakdown values reported, which are all reached well before the plateau.

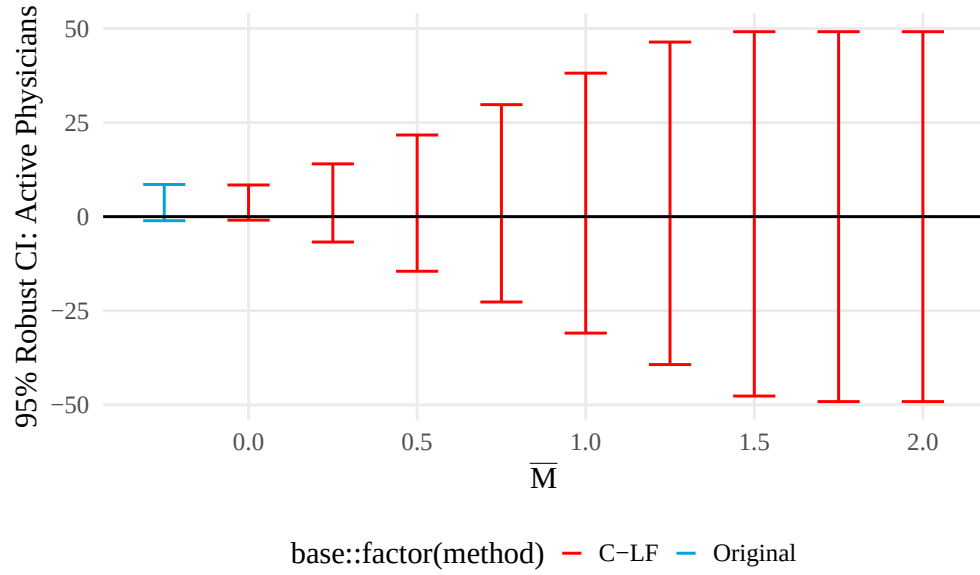


Figure I1: HonestDiD sensitivity plot for the active RHP stock (regulated states). The vertical bar at each $\bar{M}$ shows the robust 95% confidence interval; the leftmost bar ($\bar{M} = 0$) is the Original, exact-parallel-trends interval. Source: NPPES/CMS DAC.

J Bilateral Origin-Destination Flow Regression

The control-group choice in Section 4 requires that Texas RHP outflows are not disproportionately absorbed by the regulated-state donor pool; a differential surge of Texas movers specifically into control states would violate the stable unit treatment value assumption (SUTVA) implicit in treating those states as an unaffected counterfactual. We test this directly using the bilateral origin-destination framework of [Borusyak, Dix-Carneiro and Kovak \(2022\)](#).

Let Y_{odgt} denote the number of group- g physicians in origin state o who move to destination state d between quarter $t - 1$ and quarter t , per 10,000 group- g physicians in the origin state's stock at $t - 1$. We estimate

$$\begin{aligned} Y_{odgt} = & \alpha_{odg} + \mu_{ot} + \lambda_{dt} + \rho_{gt} + \beta_H [\text{RHP}_g \times \text{Post}_t \times \mathbf{1}(o = \text{TX}, d \in H)] \\ & + \beta_R [\text{RHP}_g \times \text{Post}_t \times \mathbf{1}(o = \text{TX}, d \in R)] + \varepsilon_{odgt}, \end{aligned} \quad (3)$$

where H is the set of regulated states and R is the set of less regulated states (Texas belongs to neither). The specification includes dyad-group, origin-time, destination-time, and group-time fixed effects, separating origin-specific and destination-specific time trends from the Texas-specific treatment effect. The sample is restricted to strict interstate transitions between two active practice quarters, excluding new entrants and retirements by construction. $\beta_H + \beta_R$ is the excess post-SB-8 outflow of Texas RHPs into the regulated-state donor pool.

We estimate a second regression with the roles of origin and destination reversed, so that Texas is the destination:

$$\begin{aligned} Y_{odgt} = & \alpha_{odg} + \mu_{ot} + \lambda_{dt} + \rho_{gt} + \gamma_H [\text{RHP}_g \times \text{Post}_t \times \mathbf{1}(d = \text{TX}, o \in H)] \\ & + \gamma_R [\text{RHP}_g \times \text{Post}_t \times \mathbf{1}(d = \text{TX}, o \in R)] + \varepsilon_{odgt}, \end{aligned} \quad (4)$$

$\gamma_H + \gamma_R$ is the excess post-SB-8 inflow of RHPs into Texas from the regulated and less

regulated donor pools.²⁰

With a single treated origin (or destination), we use permutation inference: for each of the 50 non-Texas states in turn, we relabel that state as the treated origin (for equation 3) or treated destination (for equation 4), re-estimate the identical regression, and compare Texas’s actual coefficient to the resulting empirical distribution of placebo estimates.

The key number for the SUTVA question is $\hat{\beta}_H$: the excess post-SB-8 flow specifically into the regulated-state donor pool, the only destination bucket that could actually contaminate our control group. $\hat{\beta}_H = 0.1118$ (permutation $p = 0.569$) is not detectably different from the outflow generated when a randomly selected other state is relabeled as the treated origin: Texas’s post-SB-8 outflow into the regulated states is not unusual relative to ordinary state-to-state variation. For context, $\hat{\beta}_R = 0.0889$ (permutation $p = 0.588$) is the analogous excess flow into the less-regulated states (not part of our control group, so not itself a SUTVA threat), and $\hat{\beta}_H + \hat{\beta}_R = 0.2008$ (permutation $p = 0.471$) is the excess flow into all other states combined, a broader quantity than what the SUTVA check actually requires, included here only as a robustness cross-check on $\hat{\beta}_H$.

For the inflow specification, Texas received more physicians from other regulated states than would be expected after SB-8’s signing ($\hat{\gamma}_H + \hat{\gamma}_R = 1.2134$, permutation $p = 0.020$). Assuming this is not a false positive (which is possible, given the number of comparisons made in this appendix), this would mean physicians moved toward Texas from other regulated states at a higher rate after SB-8 than before.

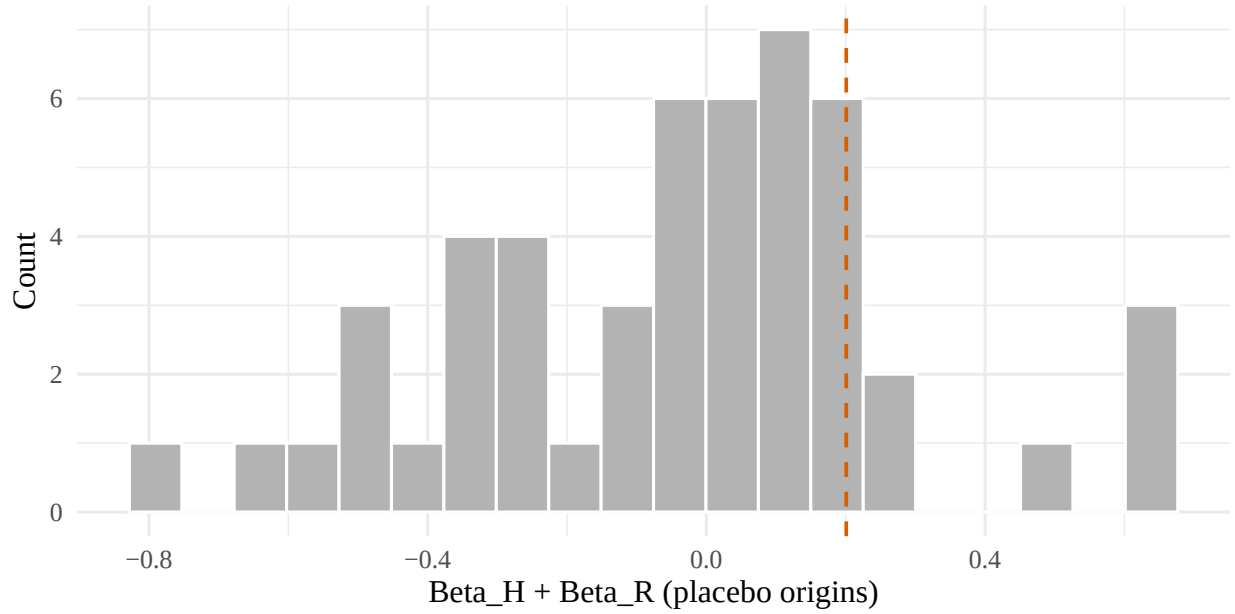
Figure J1 plots the placebo distributions against Texas’s actual estimates for both directions; the outflow panel ($\hat{\beta}_H + \hat{\beta}_R$) sits within the bulk of the placebo distribution, while the inflow panel ($\hat{\gamma}_H + \hat{\gamma}_R$) sits nearer the tail, consistent with the p-values above. This is a static, single-number version of a question that Figure 6’s leaver-share panel already asks with a time dimension: whether the share of Texas RHP leavers landing in a regulated

²⁰Because Y_{odgt} is normalized by the *origin* state’s stock in both equations, γ_H and γ_R measure the share of each sending state’s RHP stock that moved to Texas, not the share of Texas’s own RHP stock that arrived from elsewhere. This differs from the normalization underlying `enter.state_inpractice_per100k` in Table A1, which is expressed relative to Texas’s own population.

state moved specifically around SB-8’s signing. The two checks are complementary: the event-study panel shows *when* (or whether) any shift occurred quarter by quarter, while the permutation test here shows *how unusual* Texas’s post-SB-8 flow into the donor pool is relative to the same statistic computed for every other state. Neither shows evidence of the donor-pool contamination that would threaten the validity of the regulated-state control group; the inflow signal is a compositional finding about origins, not a SUTVA violation, since the SUTVA-relevant quantity is $\hat{\beta}_H$ (outflow into the donor pool) above.

Placebo distribution vs. actual TX estimate

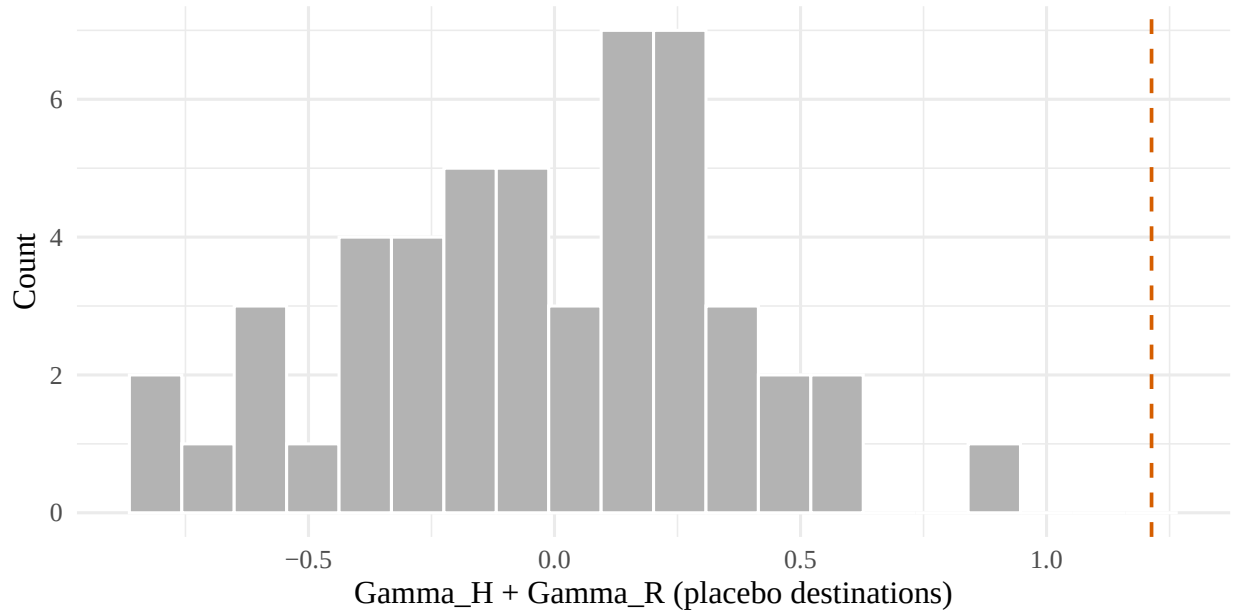
Actual TX $\beta_H + \beta_R = 0.2008$ (permutation $p = 0.471$)



(a) Outflow: $\beta_H + \beta_R$ (Texas as origin).

Placebo distribution vs. actual TX estimate (inflow)

Actual TX $\gamma_H + \gamma_R = 1.2134$ (permutation $p = 0.020$)



(b) Inflow: $\gamma_H + \gamma_R$ (Texas as destination).

Figure J1: Permutation distributions for the bilateral origin-destination flow regression. Each histogram shows the distribution of $\hat{\beta}_H + \hat{\beta}_R$ (panel a) or $\hat{\gamma}_H + \hat{\gamma}_R$ (panel b) across 50 placebo relabelings of the treated origin or destination. The dashed orange line marks Texas's actual estimate. Source: NPPES/CMS DAC.

K Leave-One-Out on Control States

Because the design has a single treated unit (Texas) against a pool of 25 regulated control states, we check that the SDID estimate is not disproportionately driven by any one control state’s trajectory. We sequentially exclude each of the 25 regulated states in turn, refit SDID on the remaining 24 (Texas as the single treated unit), and report the resulting range across all 25 leave-one-out estimates alongside how many are individually significant at the 5% level (placebo-variance SE, $B = 1,000$, each refit on its own 24-state donor pool). Table K1 reports the migration outcomes; Table K2 reports the Wagescape outcomes. No migration or wage outcome changes sign in a way that suggests a single-state dependency, and no leave-one-out estimate reaches significance for any outcome except Retirements, where every point estimate (full-sample and all 25 leave-one-out refits) carries an implausibly small placebo-variance SE (≈ 0.0001); we read this as the same measurement-degeneracy issue already flagged for Retirements elsewhere (Figure H1’s caption, Appendix I) rather than a genuine single-state dependency, since the underlying full-sample point estimate itself is economically negligible.

Leave-One-Out on Control States: Migration Outcomes (SDID)

Outcome	Full-Sample SDID $\hat{\tau}$	(SE)	Leave-One-Out Range	# Sig. at 5%
Active	−2.877	(5.729)	[−3.643, −1.173]	0/25
Leaver Share	0.049	(0.092)	[0.012, 0.062]	0/25
Entrant Share	0.044	(0.144)	[0.024, 0.062]	0/25
New Entrances	0.000	(0.012)	[−0.005, 0.005]	0/25
Retirements	−0.003	(0.000)	[−0.009, 0.000]	23/25

Table K1: Full-sample SDID $\hat{\tau}$ (all 25 regulated donor states) versus the range of $\hat{\tau}$ across 25 leave-one-out re-estimates (each excluding one regulated donor state in turn, refit independently with its own placebo-variance SE). Active, New Entrances, and Retirements are per 100,000 women of childbearing age; Leaver/Entrant Share are shares (0–1) of interstate movers. Retirements’ near-zero SEs reflect a known measurement-degeneracy issue, not precision (see main text above). Source: NPPES/CMS DAC.

Leave-One-Out on Control States: Wagescape Outcomes

Outcome	Full-Sample β_1	(SE)	Leave-One-Out Range	# Sig. at 5%
Log Salary	-0.034	(0.138)	$[-0.051, -0.021]$	0/25
Postings per 100k	-2.702	(16.002)	$[-4.296, -1.256]$	0/25
COL Salary (USD)	332.1	(22666.6)	$[-2594.5, 3128.0]$	0/25

Table K2: Full-sample β_1 (all 25 regulated controls) versus the range of β_1 across 25 leave-one-out re-estimates (each excluding one regulated control state in turn). Same specification and [Ferman and Pinto \(2019\)](#) SEs as Table A2. Source: Wagescape via Dewey Data Platform.

L Balanced-Panel Check

The NPI panel is unbalanced: physicians appear and disappear from the NPPES registry for reasons unrelated to migration (e.g., temporary lapses in registration, deactivation and later reactivation), so the active RHP stock measured on the full panel could in principle be driven by this compositional churn rather than by genuine state-to-state migration. As a check, we restrict to the 1,078,999 NPIs (65.8% of the panel) that are active in every quarter of the main estimation window (2015Q2–2022Q2) and re-fit the SDID specification of Section 5.1 to this restricted sample for the active RHP stock. Leaver/Entrant Share, New Entrances, and Retirements are not re-estimated on this restricted sample: by construction, a physician active in every quarter of the window cannot register a career entry, a career exit, or (for the destination-share outcomes) a state-to-state move that leaves an inactive gap, so those outcomes are not well-defined here. Table L1 shows the balanced-panel SDID estimate alongside the main-sample SDID estimate for comparison; both remain null and of comparable magnitude, indicating the main result is not an artifact of registry entry/exit composition.

Balanced-Panel Check: Active RHP Stock (SDID)

Sample	SDID $\hat{\tau}$	(SE)
Main Sample (Unbalanced)	−2.877	(5.729)
Balanced Panel	−0.385	(1.811)

Table L1: SDID estimate (Texas vs. synthetic Texas) for the active RHP stock: full unbalanced NPI panel versus restricted to NPIs active in every quarter of the 2015Q2–2022Q2 estimation window, per 100,000 women of childbearing age. Placebo-variance SE ($B = 1,000$), same specification as Table 5. Source: NPPES/CMS DAC.

M Excluding General Practice, Pediatrics, and Oncology from the Control Pool

The non-RHP comparison group used throughout is defined residually (any physician not classified as OB/GYN, OB-practicing Family Medicine, or Emergency Medicine), so it includes General Practice physicians (NUCC taxonomy 208D), Pediatricians (2080), and oncologists (Hematology & Oncology, Medical Oncology, Radiation Oncology, and Surgical Oncology) by default, without any of these groups having been separately examined. As a sensitivity check, we drop 208D and 2080 from the regulated-states sample entirely (5.9% of observations) and separately extend that exclusion to the four oncology taxonomies, re-estimating β_1 for the active RHP stock under each, consistent with the main body’s Active-only DDD scope (Section 5). This check is specific to the triple-difference design: it tests whether the non-RHP comparison arm’s composition matters, a question that does not have an analogous SDID specification, since SDID operates only on the RHP-only panel and never includes these specialties regardless. Table M1 shows the result is essentially unchanged under either exclusion: it does not change sign in a substantively meaningful way, and remains statistically indistinguishable from zero.

Excluding General Practice, Pediatrics, and Oncology from the Control Pool

Outcome	Main Sample β_1	(SE)	Excl. 208D + Peds β_1	(SE)	Excl. 208D + Peds + Onc. β_1	(SE)
Active	23.020	(41.595)	19.141	(25.086)	18.814	(58.420)

Table M1: Main-sample β_1 (208D, 2080, and oncology taxonomies included in the non-RHP control pool, as throughout the main text) versus β_1 with 208D and 2080 dropped from the regulated-states sample, versus β_1 with 208D, 2080, and the four oncology taxonomies (Hematology & Oncology, Medical Oncology, Radiation Oncology, Surgical Oncology) all dropped, active RHP stock per 100,000 women of childbearing age. Same specification and Ferman and Pinto (2019) SEs as Table A1. Source: NPPES/CMS DAC.

N Placebo Specialty Test

If the main-text result reflects reproductive-care-specific exposure to SB-8 rather than a general Texas physician-market trend, a specialty with no such exposure should show no differential Texas trend post-signing. We test this using General Surgery (NUCC taxonomy prefix 2086) as a placebo group, already present in the underlying panel at the state $\times$ taxonomy $\times$ quarter level. Because general surgeons are already part of the non-RHP pool by construction, and there is no natural within-surgery treated/comparison split, we substitute General Surgery for the RHP outcome and fit the same SDID specification used throughout (Texas as the single treated unit, the regulated states as the donor pool, placebo-variance inference), rather than the DDD triple-difference spec. Table N1 shows no evidence of a differential Texas trend in general surgeon density post-SB-8.

Placebo Specialty Test: General Surgery (SDID)

Outcome	SDID $\hat{\tau}$	(SE)
General Surgery (Placebo)	−0.472	(1.367)

Table N1: SDID estimate (Texas vs. synthetic Texas) for General Surgery physicians (taxonomy 2086) substituted in place of RHP as the outcome, per 100,000 women of childbearing age, regulated-states donor pool. Placebo-variance SE ($B = 1,000$), same specification as Table 5. Source: NPPES/CMS DAC.

O Two-Way Fixed-Effects Difference-in-Differences

For comparability with the wider literature, Table O1 reports a plain two-way fixed-effects (TWFE) difference-in-differences estimate for every migration and Wagescape outcome: Texas versus the regulated states, restricted to reproductive health physicians, with unit and date fixed effects (the same collapsed panel used throughout, without the non-RHP triple-difference term). Ferman and Pinto (2019) standard errors are used throughout, as in the rest of the paper. Every outcome is statistically indistinguishable from zero, consistent with the main triple-difference results.

Two-Way Fixed-Effects DiD: Texas vs. Regulated States (RHP Only)

Outcome	β (TX $\times$ Post-SB8)	(SE)
<i>Migration (NPI)</i>		
Active	−11.855	(29.752)
Exit State	0.111	(0.955)
Enter State	0.220	(0.628)
New Entrances	0.558	(1.777)
Retirements	0.001	(0.018)
<i>Wagescape</i>		
Log Salary	0.021	(0.145)
Postings per 100k	2.711	(3.475)
COL Salary (USD)	2040.9	(22811.1)

Table O1: Two-way fixed-effects DiD estimates (Texas $\times$ Post-SB8), reproductive health physicians only, regulated-states control group. Migration outcomes per 100,000 women of childbearing age; unit (state $\times$ RHP) and date fixed effects. Wagescape outcomes: log COL-adjusted salary, job postings per 100,000 women of childbearing age, and COL-adjusted salary in USD levels; unit (state $\times$ RHP) and quarter-year fixed effects. Ferman and Pinto (2019) heteroskedasticity-corrected SEs in parentheses. Sources: NPPES/CMS DAC; Wagescape via Dewey Data Platform.

P Physician Age-Cohort Heterogeneity

The migration results above pool physicians of all career stages. Younger physicians choosing an initial practice location bear no relocation cost, while physicians nearing retirement face a lower fixed cost of early workforce exit than of interstate relocation (Section 6); either margin could in principle generate a differential migration response that the pooled estimate averages away. We use the CMS Doctors and Clinicians (DAC) file’s medical school graduation year (Section 4.3) to split reproductive health physicians into two age cohorts: physicians less than 10 years past graduation (“young”) and physicians more than 20 years past graduation (“old”); we then re-estimate the SDID specification of Section 5.1 separately within each cohort, for the active RHP stock. Physicians 10–19 years past graduation are excluded from both subsamples rather than assigned to either.

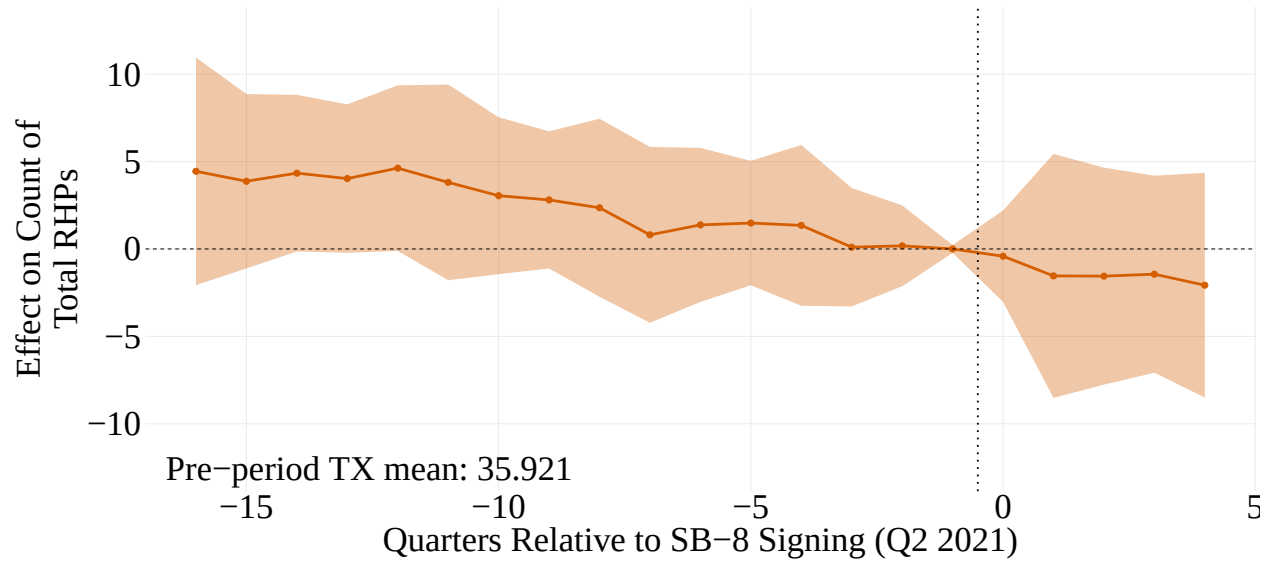
Because the DAC match is itself non-random (Section 4.3 documents a mobility skew in the matched subsample), and because splitting an already-restricted DAC-matched, RHP-only panel by age cohort further thins the data, we first confirmed that all 25 regulated states plus Texas retain complete quarterly coverage in both age-cohort subsamples over the estimation window (2017 Q2–2022 Q1, the window over which DAC coverage and the pre-*Dobbs* cutoff overlap) before fitting SDID; no donor state was dropped from either subsample.

Table P1 reports the results. Both cohorts show a small SDID point estimate relative to its standard error, and neither is statistically distinguishable from zero. The two point estimates are not meaningfully different from each other given the size of their standard errors. Figure P1 shows the corresponding per-quarter SDID event studies, following the same Ciccio (2024) construction used throughout the paper: neither cohort shows a sustained pre-period trend or a post-signing break.

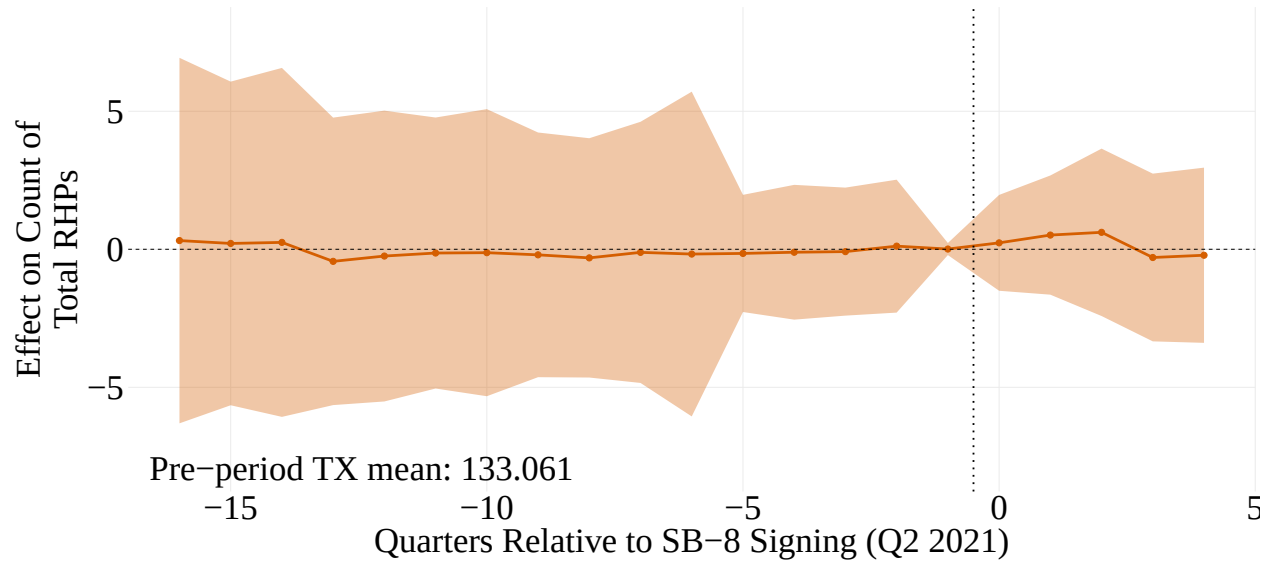
SDID by Physician Age Cohort: Active RHP Stock

Age Group	SDID $\hat{\tau}$	SDID SE	SC $\hat{\tau}$	SC SE
Young (<10 yrs since grad.)	-1.405	2.574	-0.525	3.600
Old (>20 yrs since grad.)	0.170	1.173	3.583	6.689
Donor states (regulated)		25		
Pre-periods		16		

Table P1: SDID and SC estimates for the active RHP stock, fit separately on physicians less than 10 years past medical school graduation (“Young”) and more than 20 years past graduation (“Old”), per 100,000 women of childbearing age. Regulated-states donor pool; placebo-variance SEs ($B = 1,000$). Source: NPPES/CMS DAC.



(a) Young (less than 10 years since graduation).

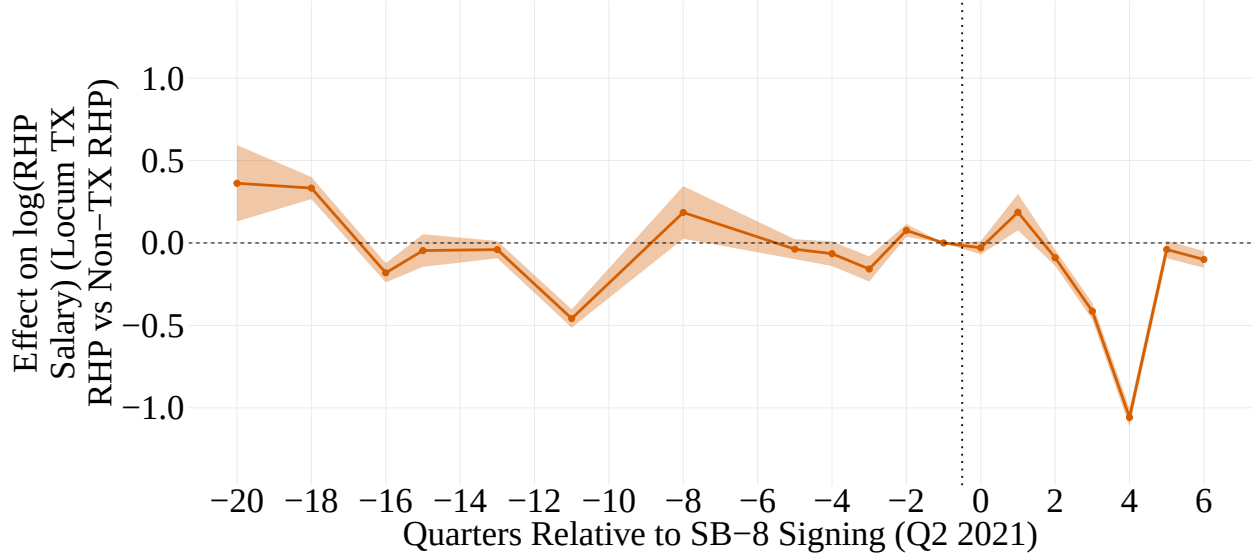


(b) Old (more than 20 years since graduation).

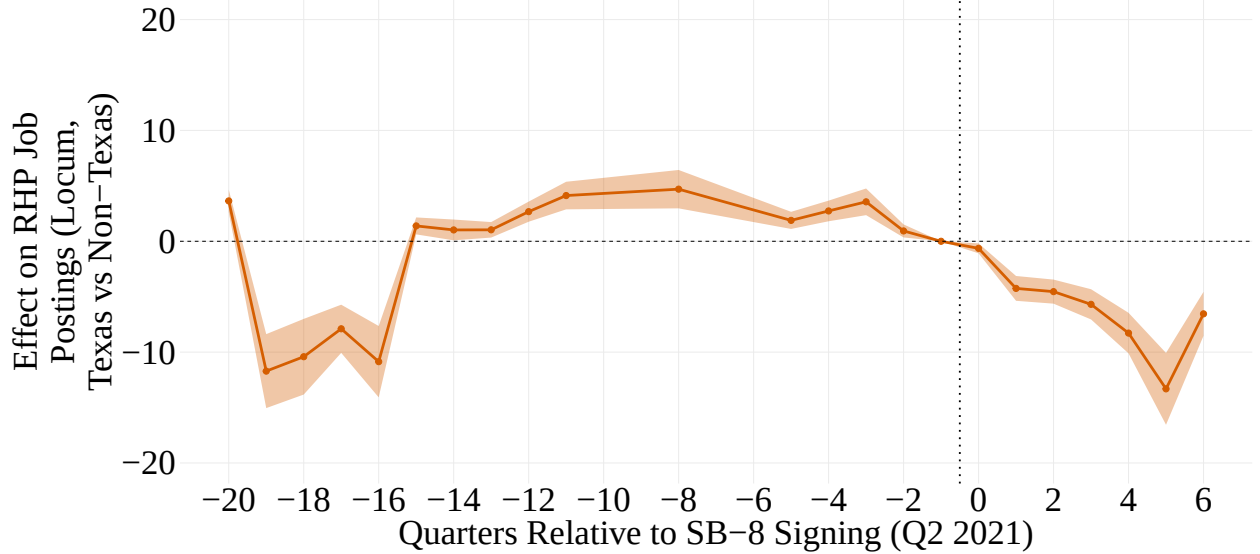
Figure P1: Per-quarter SDID treatment effect on the active RHP stock (Texas versus synthetic Texas, net of the time-weighted pre-period baseline), fit separately by physician age cohort, following [Ciccia \(2024\)](#). Shaded regions show 95% confidence intervals from a leave-one-out placebo procedure over the regulated-states donor pool. Quarters relative to SB-8 signing (May 19, 2021); event time 0 is Q2 2021. Source: NPPES/CMS DAC.

Q Locum Tenens Subsample: Event Studies

Figure [Q1](#) shows the triple-difference event study analogs to the locum tenens DDD estimates in Table [A2](#), Panel C (Appendix [A](#)).



(a) Log posted salary: Texas RHP locum vs. non-Texas RHP locum differential, quarterly.

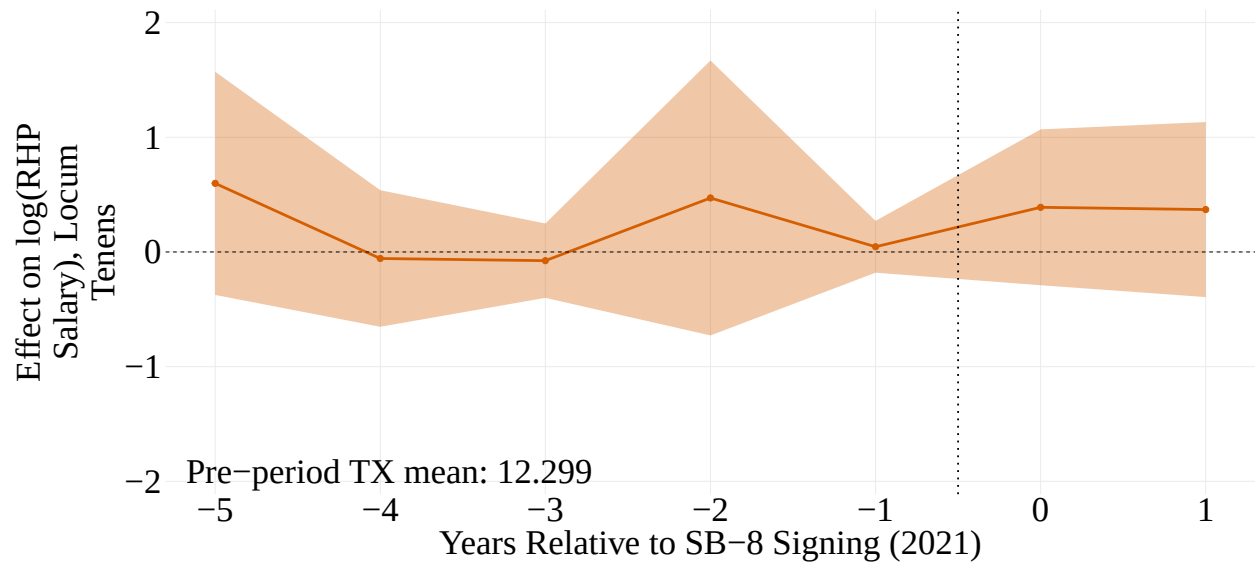


(b) Locum RHP postings per 100k women: Texas vs. non-Texas differential, quarterly.

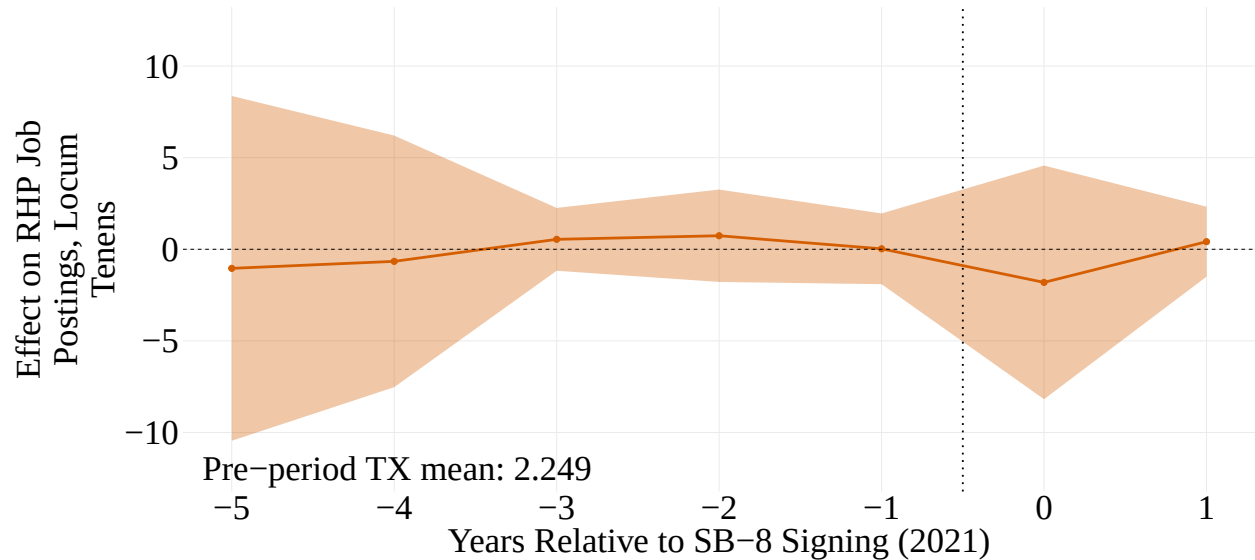
Figure Q1: Triple-difference event study estimates for the locum tenens subsample. Coefficients β_k are the quarterly treatment effects ($\text{TX} \times \text{RHP} \times \text{Post}$) relative to Q1 2021 (reference period: event time -1). Panel (a) uses log BEA RPP cost-of-living adjusted posted salary; panel (b) uses locum RHP job postings per 100,000 women of childbearing age. State, year, and quarter fixed effects are included. Sample: Wagescape physician job postings with “locum” in the job title, 2016–2022, all U.S. states and D.C. Shaded regions show 95% confidence intervals. Cluster-robust SEs clustered by state. Source: Wagescape via Dewey Data Platform.

We do not report an SDID event study for the locum subsample at the paper’s usual quarterly granularity. Restricting to locum-tenens postings, further restricted to reproductive health providers (and to a disclosed salary, for the salary outcome), leaves a panel too

sparse for SDID’s balanced-panel requirement at that frequency: zero of the 25 regulated states, including Texas itself, have a locum RHP posting in every quarter of the estimation window, for either outcome. Aggregating to annual frequency restores a usable donor pool (Figure Q2): 8 donor states for log salary, 13 for posting volume, both alongside Texas as the treated unit. 2021 is coded as the first post-period year, since SB-8 was signed in Q2 2021. Pre-period coefficients are noisy at this coarser frequency, expected given the small number of pre-period years (5) and the thinness of the underlying sample, but no systematic pre-trend is evident, and neither post-period year (2021, 2022) is statistically distinguishable from zero for either outcome, consistent with the quarterly DDD result above.



(a) Log posted salary, locum tenens RHP.



(b) RHP job postings per 100k women, locum tenens.

Figure Q2: Per-year SDID treatment effect for the locum tenens subsample (Texas versus synthetic Texas, net of the time-weighted pre-period baseline), following [Ciccia \(2024\)](#), aggregated to annual frequency to obtain a usable donor pool (see text). Shaded regions show 95% confidence intervals from a leave-one-out placebo procedure over the regulated-states donor pool. Years relative to SB-8 signing (May 19, 2021); 2021 is the first post-period year. Source: Wagescape via Dewey Data Platform.

R Full-Sample (All-States) Control Group

The main text uses [Dench et al. \(2025\)](#)’s “regulated states” (Section 4) as the primary control group, since Table 4 shows they are a closer pre-period match to Texas than the full pool of states. As a robustness check on this choice, Table R1 reports SDID and SC estimates for all eight outcomes using the full 50-state donor pool (all states except Texas) instead. Results are qualitatively unchanged: every outcome remains statistically indistinguishable from zero under both control-group definitions.

SDID and SC Estimates: All-States Control Group

Outcome	SDID	SDID SE	SC	SC SE
<i>Panel A: NPI Migration Outcomes</i>				
Count of Total RHPs	-2.784	(6.336)	-0.252	(13.043)
Leaver Share to Regulated	0.010	(0.099)	0.015	(0.095)
Entrant Share from Regulated	0.045	(0.084)	0.036	(0.086)
New Entrances	-0.002	(0.007)	0.010	(0.014)
Retirements	-0.000	(0.003)	0.000	(0.001)
<i>Panel B: Wagescape Wage Outcomes</i>				
log(RHP Salary)	0.061	(0.178)	0.023	(0.133)
RHP Job Postings	1.620	(3.522)	2.309	(4.095)
RHP Salary (USD)	3,504	(20,061)	-2,928	(12,484)

Placebo variance ($B = 1,000$). Count of Total RHPs, New Entrances, and Retirements are per 100,000 women of childbearing age; Leaver/Entrant Share are shares (0–1) of interstate movers landing in or arriving from a regulated state. SDID and SC operate on the RHP-only panel with Texas as the single treated unit and the full 50-state donor pool (all states except Texas).

Table R1: SDID and SC estimates for all eight outcomes, all-states (full 50-state) donor pool, compare to Table D1’s regulated-states estimates. SDID and SC operate on the RHP-only panel with Texas as the single treated unit; inference via placebo variance ($B = 1,000$). DDD is omitted here since it is not separately re-estimated on the all-states control group in this paper (Table A1’s DDD estimates use the regulated-states control group only). NPI outcomes are per 100,000 women of childbearing age. Sources: NPPES/CMS DAC; Wagescape via Dewey Data Platform.